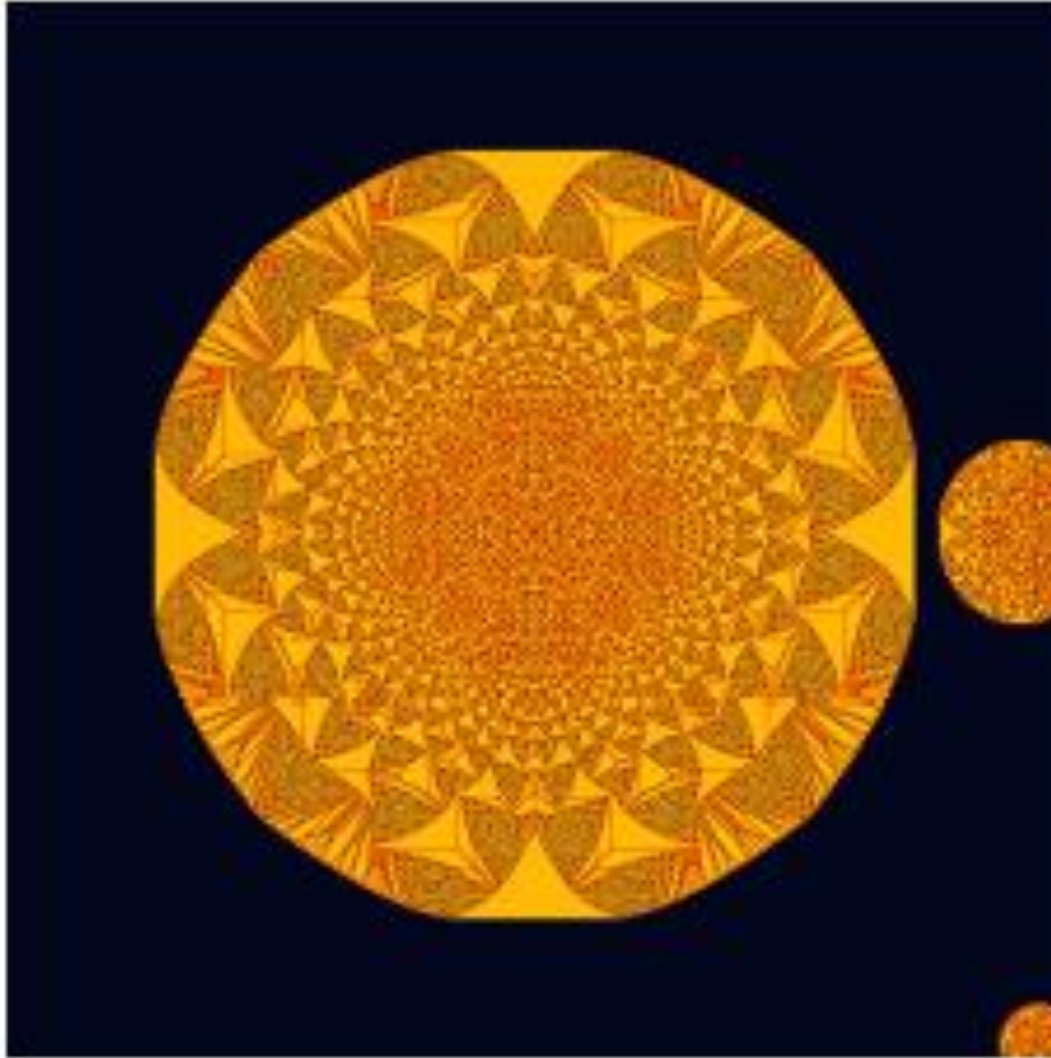
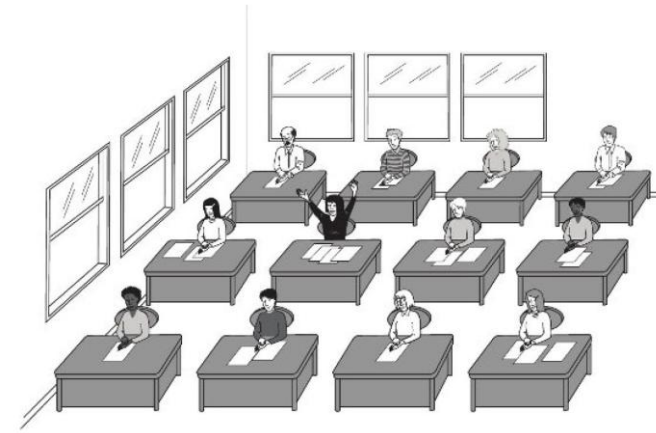


Recap: Sandpiles, relation between Source and Sink



After addition
of 2^{18} grains
in same point:

Center driven



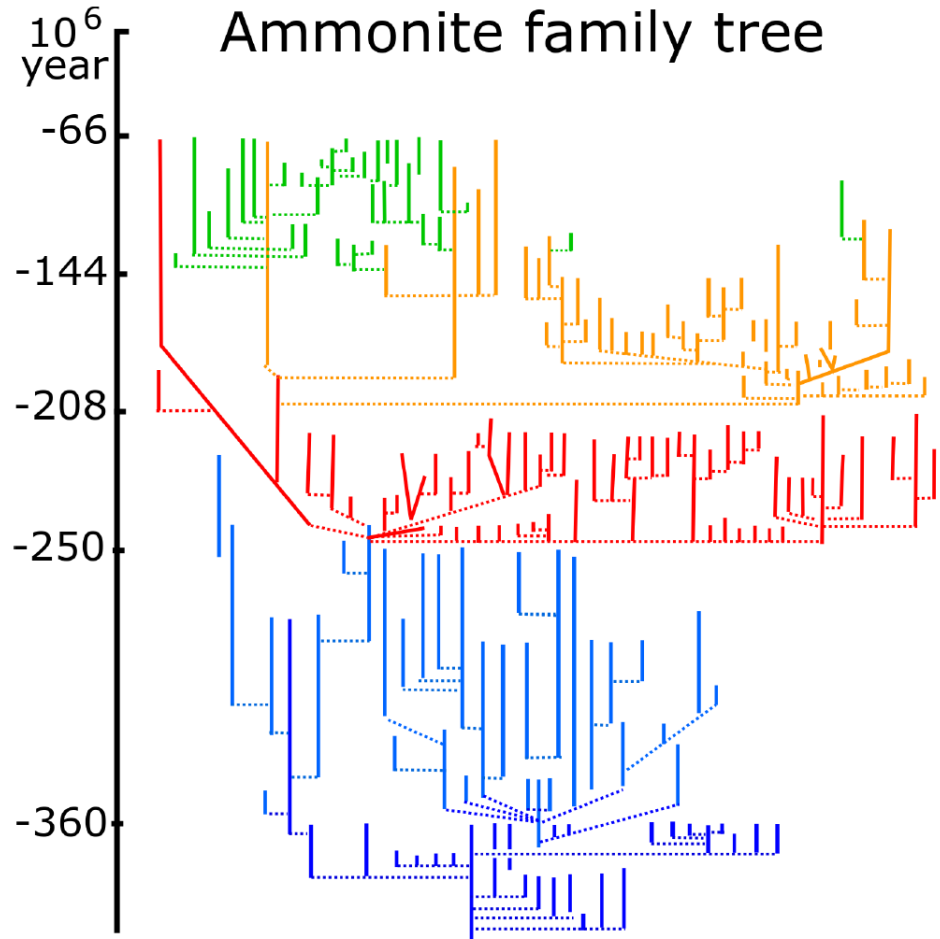
Boundary driven

$$L \sim \int_0^\infty s \cdot P(s) ds \sim \int_0^{L^D} s^{1-\tau} ds = L^{D \cdot (2-\tau)}$$
$$\Rightarrow 1 = D \cdot (2 - \tau) \Rightarrow \tau = 2 - \frac{1}{D}.$$

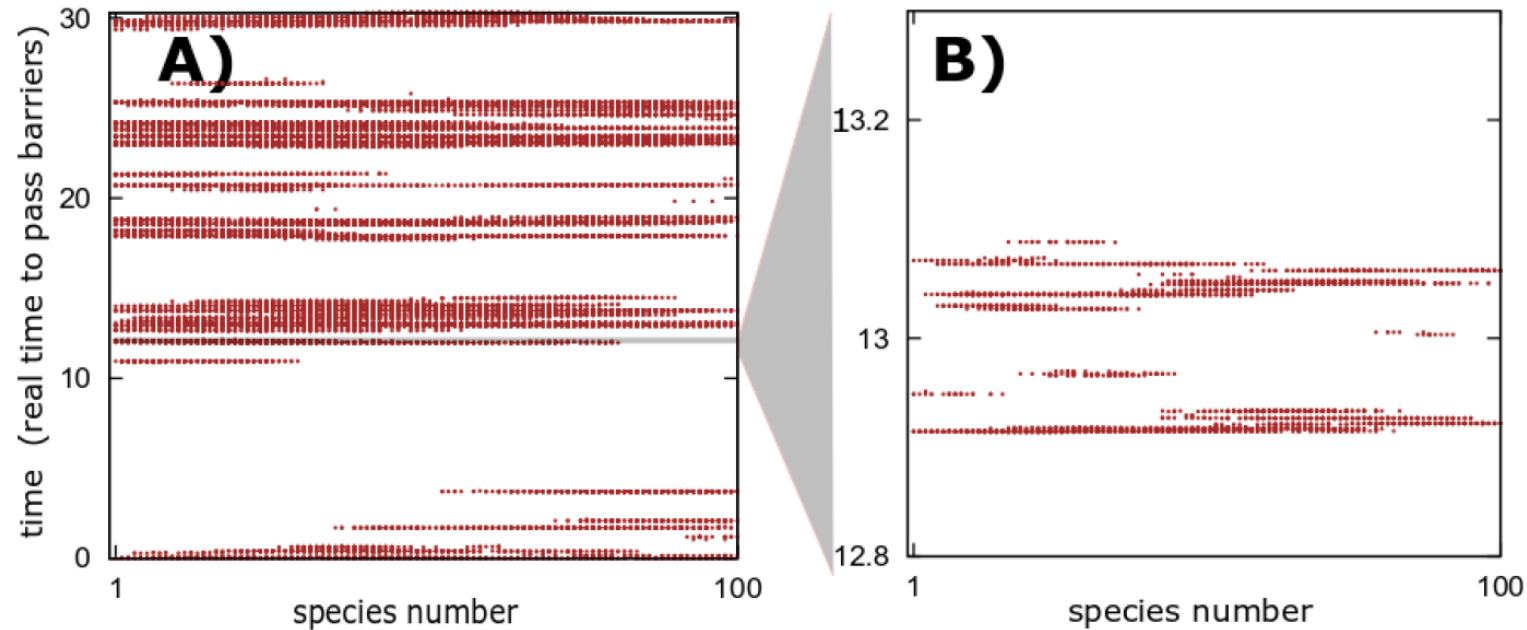
Corner driven

Recap: Extremal dynamics, finding the extreme

Ammonite family tree



- At each time step of size dt :
Select each of the $\{B_i\}_{i=1,N}$ with probability $\propto e^{-B_i/\mu} dt$. This selection defines a list of active sites. For members in this list, replace them as well as their nearest neighbors by new random numbers in $[0, 1]$.



Time

Lectures in Stochastic Simulations & Agent based models

Kim Sneppen, September 2025

Stochastic models...

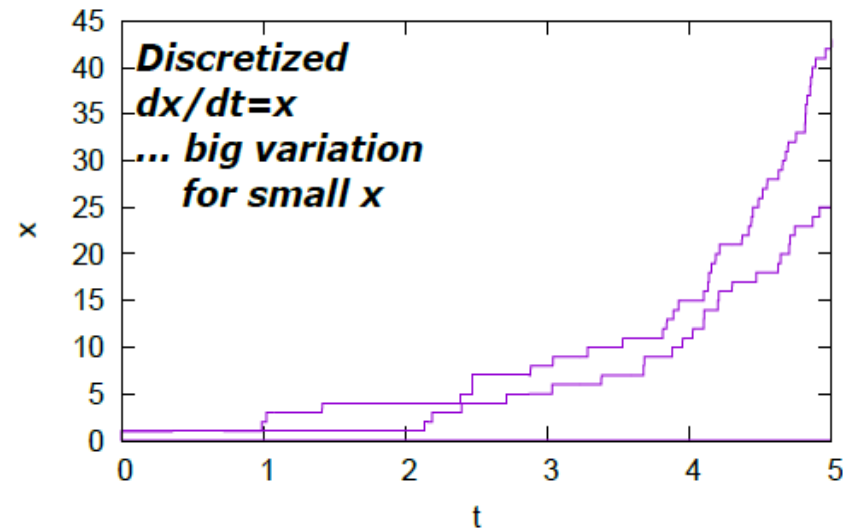
Discrete variable $x=0,1,2,3,\dots$

- $dx/dt=x,\dots$ Could be an exponential (branching) process... , x units that each replicates with rate 1. Simulate stochastic, starting with $x=1$

$$\frac{d\langle x \rangle}{dt} = \langle x \rangle$$

for small Δt can understand $x\Delta t$ as a probability that one add 1 unit

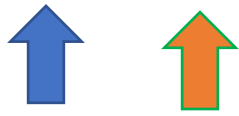
- $x \rightarrow x + 1$ with probability $x\Delta t$



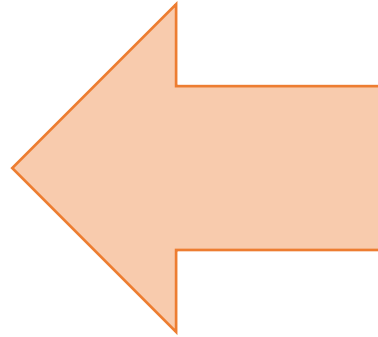
Different
Realizations →
Different outcomes
(for ex. between
Covid-19 in
different countries)

Birth & death ... *here (nearly) like critical branching*

$$\frac{d\langle x \rangle}{dt} = \langle x \rangle - \langle x \rangle$$



2 Independent processes



$$dx = \text{sqrt}(2x) * dR$$

R=Random walk, $dR^2 = dt$, $\langle (R(t) - R(t'))^2 \rangle = t - t'$.

Or in detail (the Ito way):

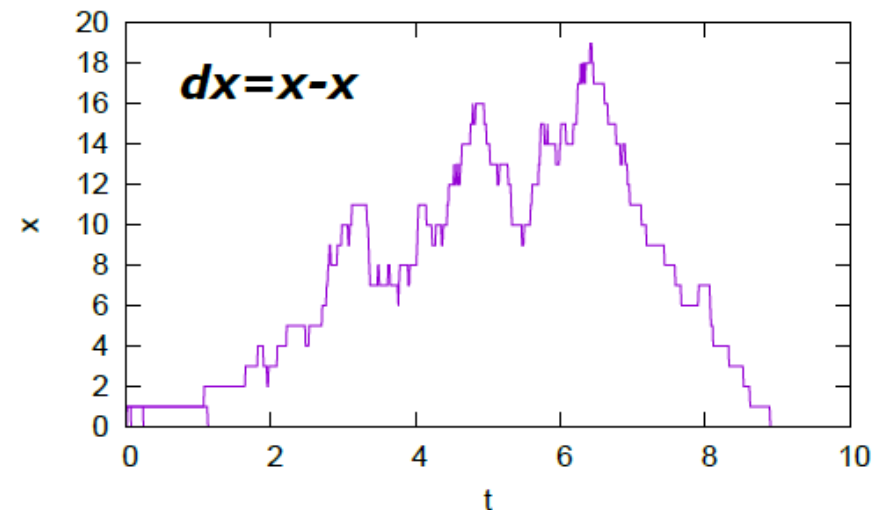
$$dx(t) = x(t+dt) - x(t) = \text{sqrt}(2x(t)) * [R(t+dt) - R(t)]$$

... notice variance is $2*x$, the 2 because of 2 processes

for small Δt can understand $x\Delta t$ as a probability that one add 1 unit

- $x \rightarrow x + 1$ with probability $x\Delta t$
- $x \rightarrow x - 1$ with probability $x\Delta t$

(How do you think first return Probability scales?)



Stochastic models (production and decay)

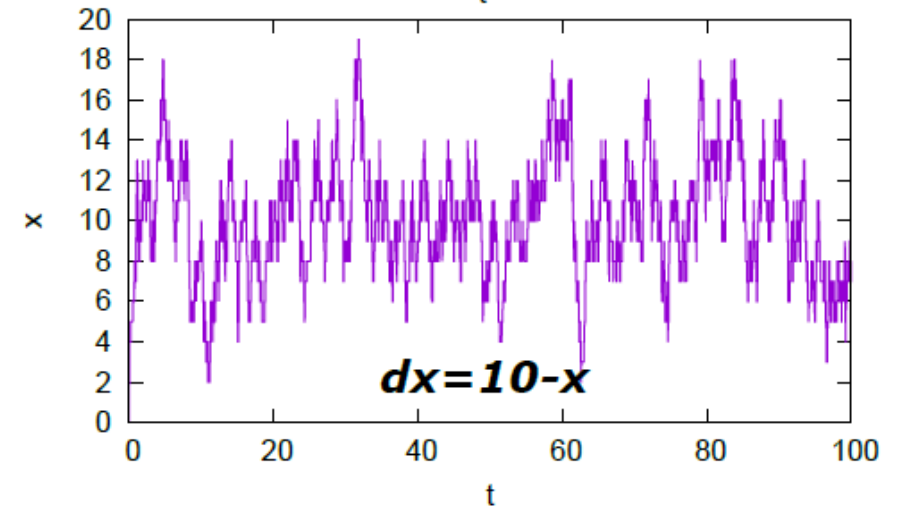
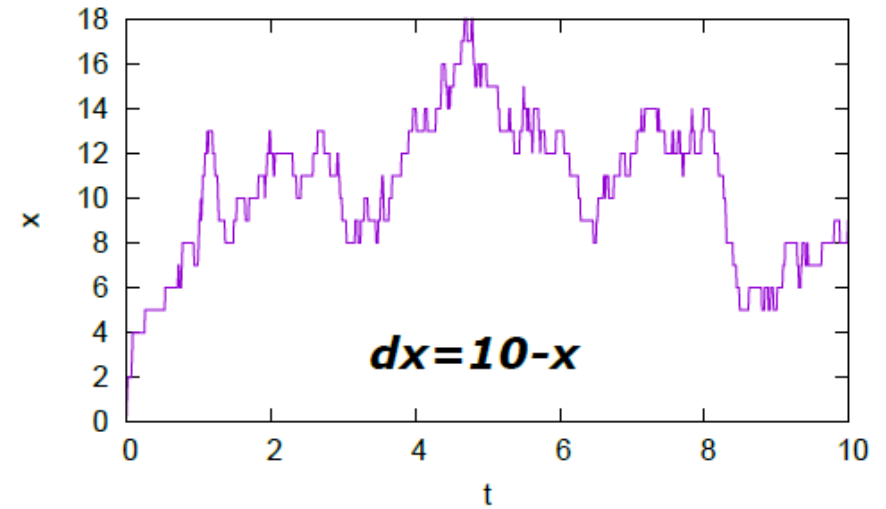
$$\frac{dx}{dt} = Q - \frac{x}{\tau}$$

$$x(t + \Delta t) = x(t) + Q \cdot \Delta t - \frac{x}{\tau} \cdot \Delta t$$

for small Δt can rephrase

- $x \rightarrow x + 1$ with probability $Q\Delta t$
- $x \rightarrow x - 1$ with probability $x\Delta t/\tau$

Must fulfill: $Q\Delta t \ll 1$ and $x\Delta t/\tau \ll 1$.



Gillespie method=event-based simulation

- Only do stuff when something is changing... jump to events;
- → calculate times when something is happening next:

A random event happens with rate r_i , then:

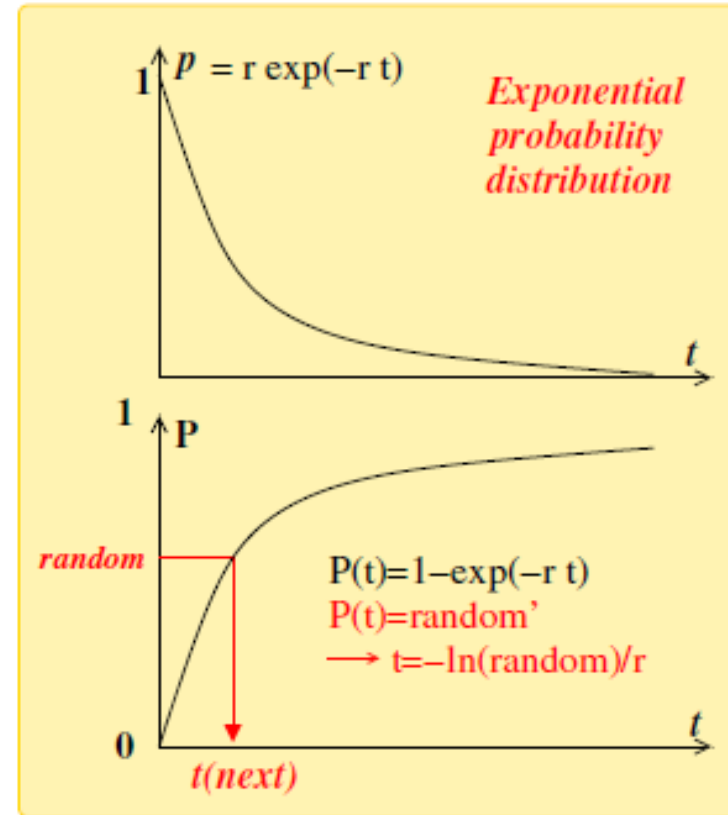
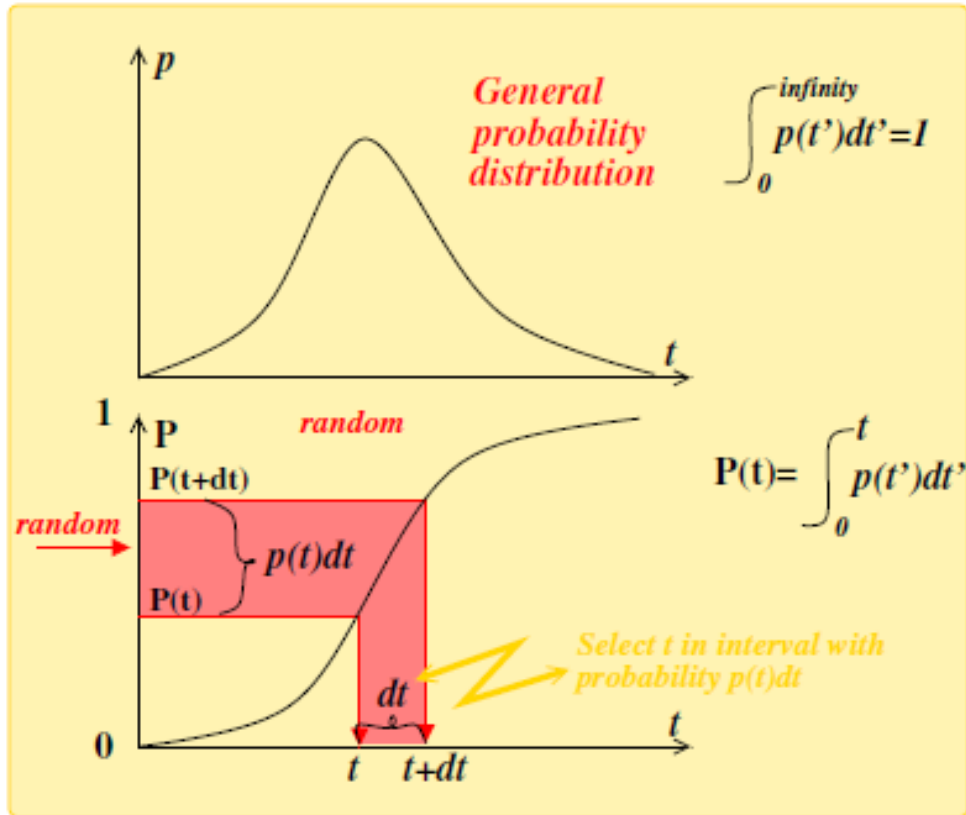
$$P_{next-event}(after \Delta t) = \exp(-r_i \Delta t)$$

Next event then happens at time:

$$\Delta t_i = -\frac{1}{r_i} \cdot \ln(ran_i)$$

Selecting a number from a distribution...

go to cumulative distribution and invert a random number:



Two competing events: (production Q and decay x/τ)

$$\frac{dx}{dt} = Q - \frac{x}{\tau}$$

- Select time step for increase in x , $\Delta t_Q = -\ln(\text{ran}_1)/Q$ by choosing a random number $\text{ran}_1 \in [0, 1]$.
- Select time step for decrease in x , $\Delta t_x = -\ln(\text{ran}_2)\tau/x$ by choosing a random number $\text{ran}_2 \in [0, 1]$.
- If $\Delta t_Q < \Delta t_x$ then update $x = x + 1$ and set time counter $t = t + \Delta t_Q$.
- If $\Delta t_Q > \Delta t_x$ then update $x = x - 1$ and set time counter $t = t + \Delta t_x$.

When calculating averages remember:

$$\langle x \rangle = \sum x(i-1) \Delta t(i) / \sum \Delta t(i)$$

Where i is timestep of varying length

The key assumption is that we consider systems without memory. That implies that any event only depends on the quantified state of the system, and that this occurrence is independent on how much time has passed since there last was a change in the system.

Mini tutorial: How could the event sizes be larger than one unit in a system?

Mini tutorial: If event size is doubled for one of the processes, what would happen to the stochastic development?

**Several processes, $i=1,2,3,\dots$ each associated with
different changed Δ_i**

- 1 Monte Carlo step: Generate random numbers ran_i to determine the time-step $\Delta t_i = -\frac{\Delta_i}{r} \cdot \ln(ran_i)$ for all potential events i , and select the first event for updating.
- 2 Update: Increase the time by the generated time-step (from Step 1) and update the variable x with the change Δ_i associated to the selected event.

Basic assumption:

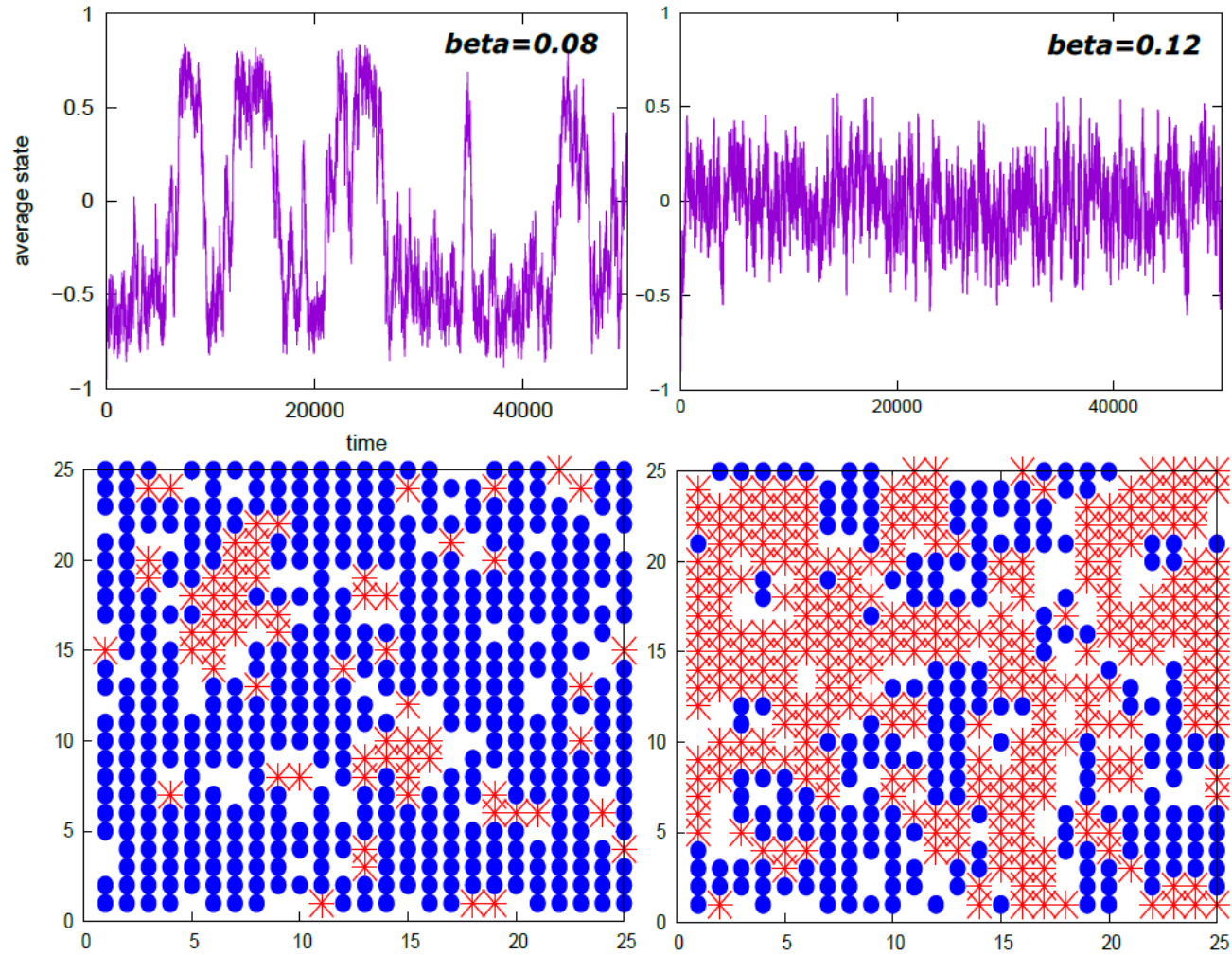
- Independent Poisson processes,... no memory,... Is ok to do one change, and then update all times from new random numbers (disregard old times equivalent to update to new one if rate not changed)
- E.g. this would not be true for simulating for example death among siblings. Their death would be correlated because of their similar age, i.e. ageing of components in the system

Ising model analog for social system modeling:

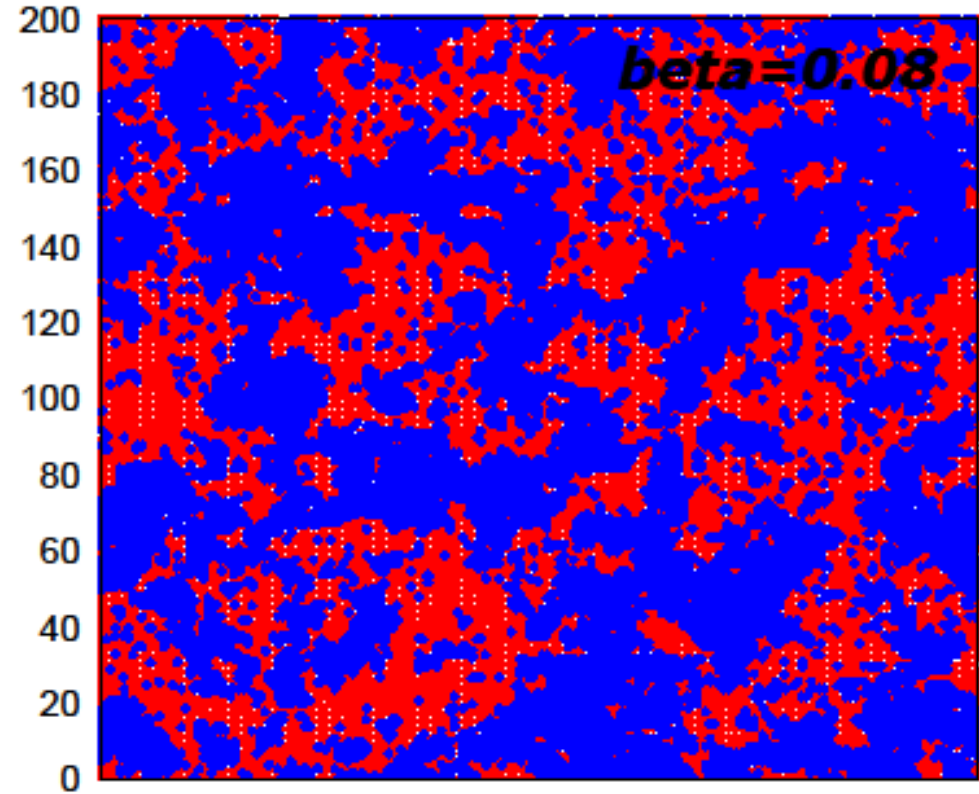
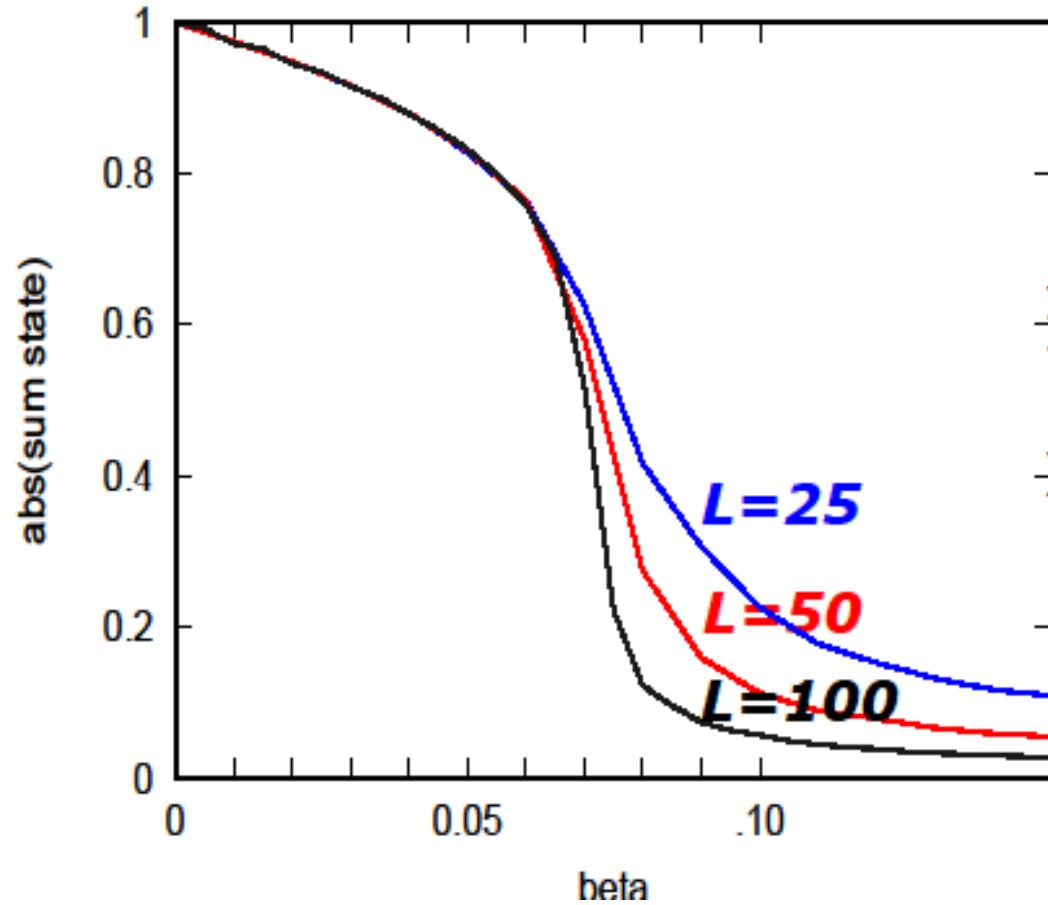
- Chose a site (x,y) . If the state s of this site is either $+1$ or -1 one randomly chose one of its 4 neighbor sites. The state of this neighbor site is adjusted towards s . Thus if $s = +1$ the neighbor change from 0 to 1 or from -1 to 0. Thus if $s = -1$ the neighbor change from 0 to -1 or from 1 to 0. If the two sites have same state value, or if $s = 0$ then nothing happens
- with probability β one select a random site and assign it a new random state, -1 , 0 or 1 with equal probability.



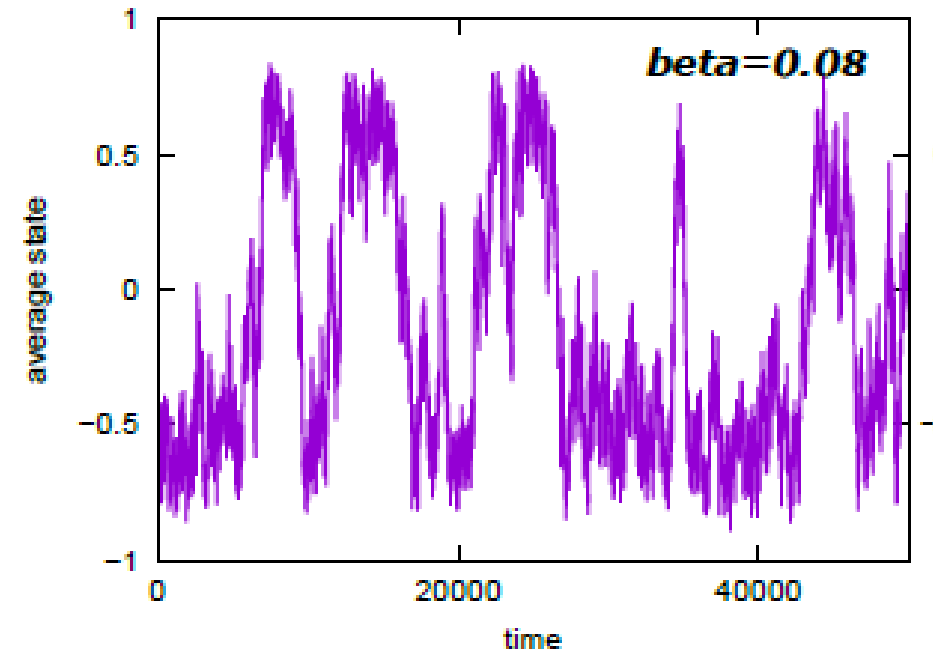
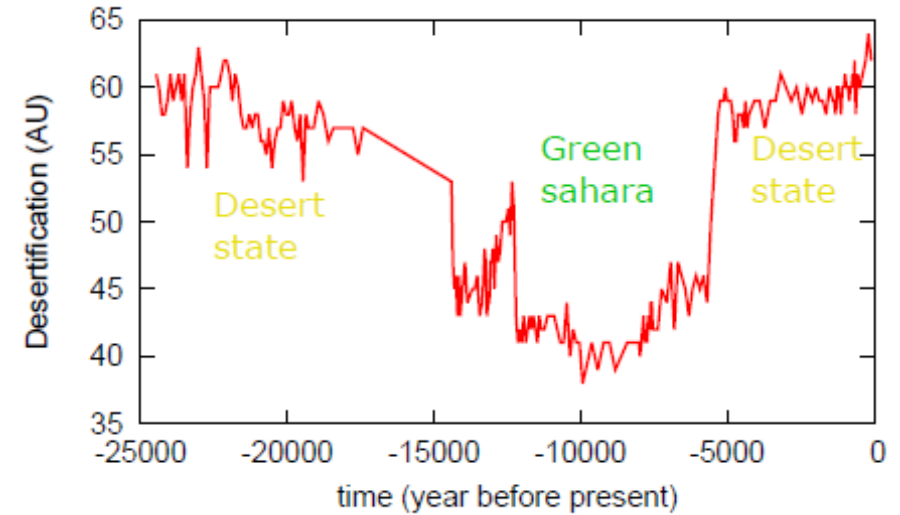
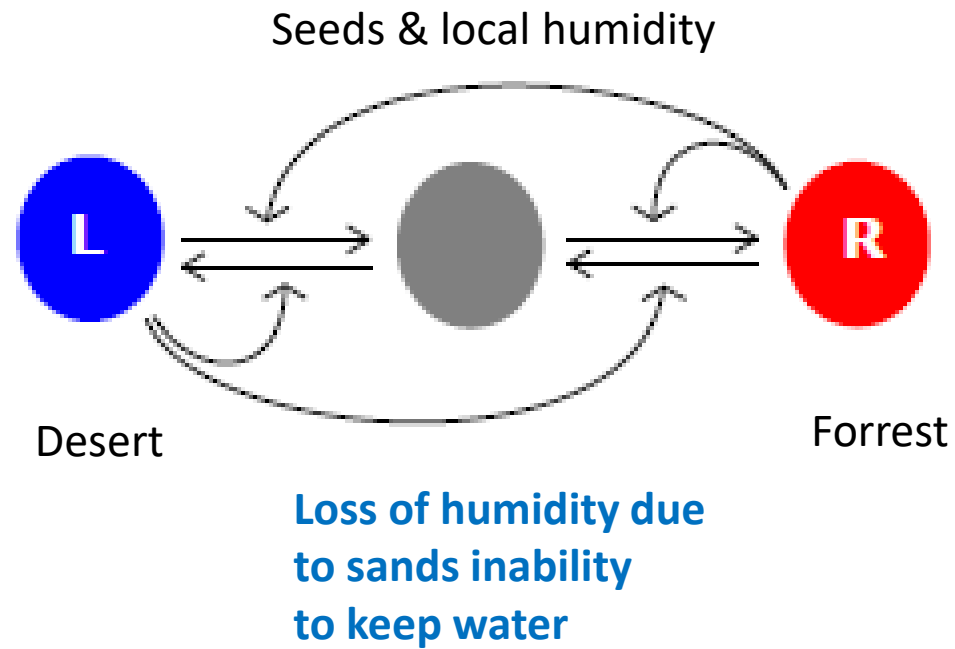
Dynamics & Typical states:



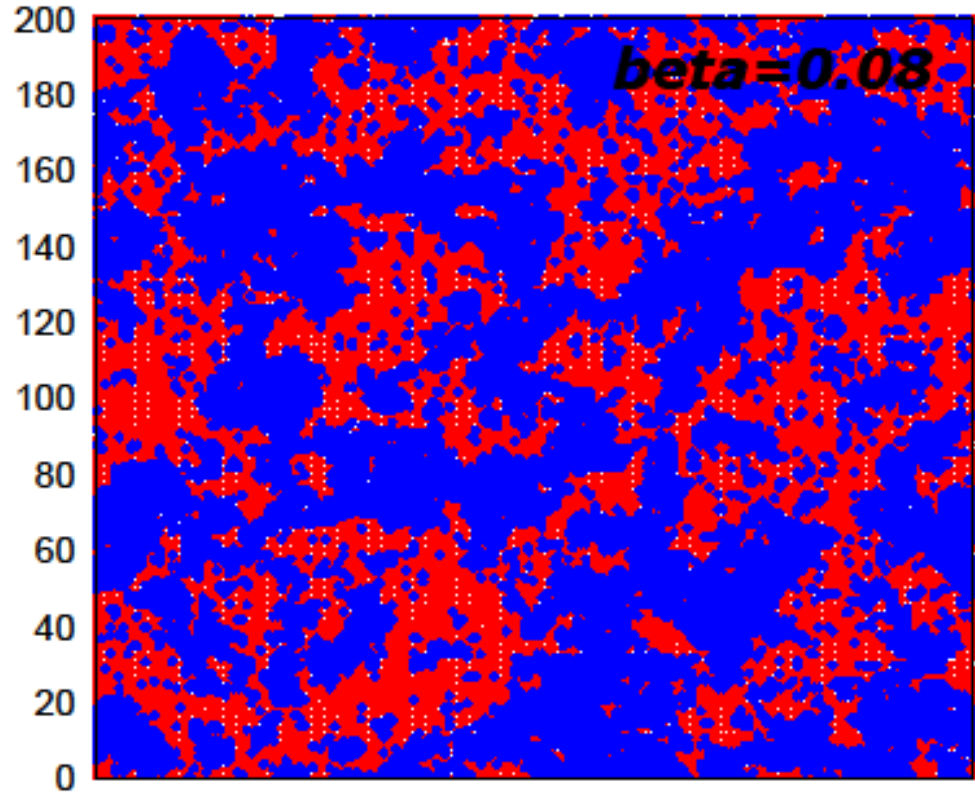
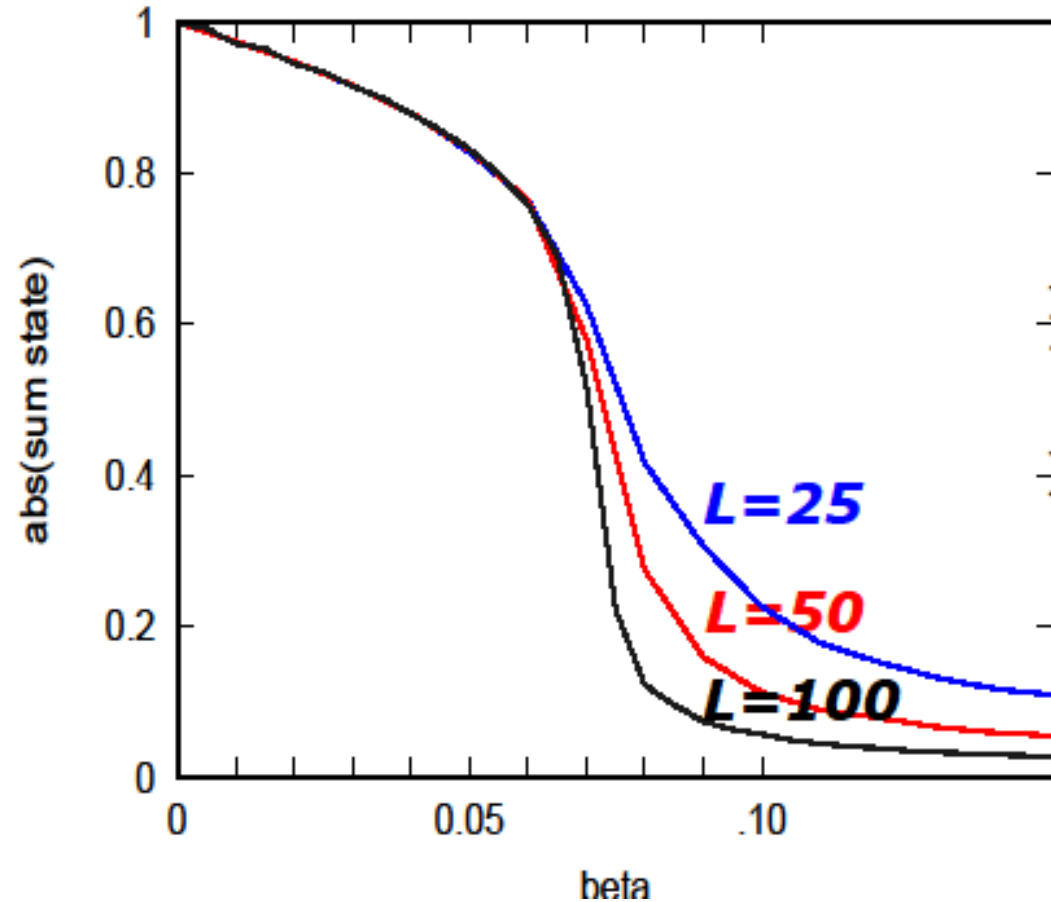
“Opinion dominance” (Magnetization):



A possible application:

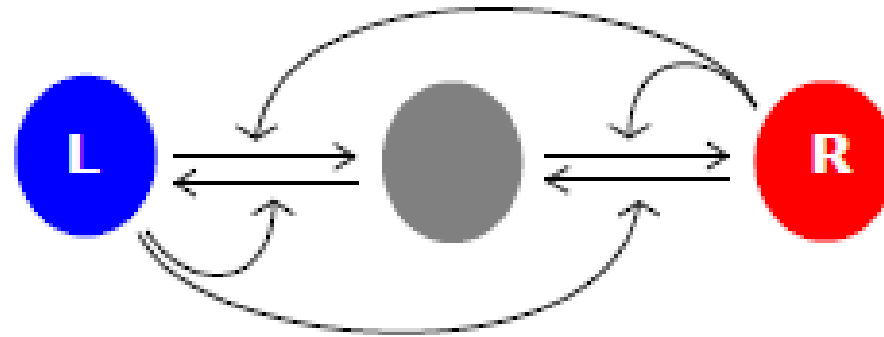


“Opinion dominance” (Magnetization):

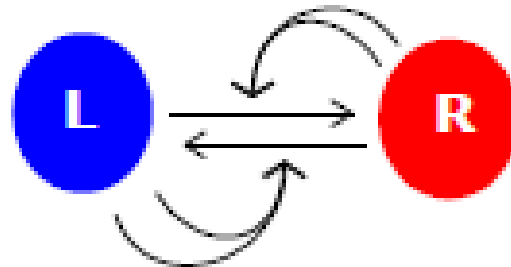


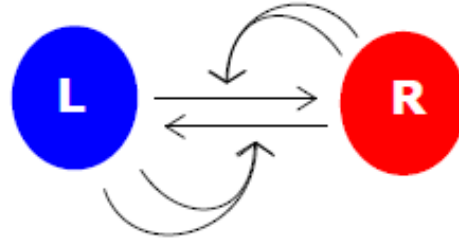
Simplifying the model: No geometry & 2 state:

3 state similar to 2-state+cooperativity



To go from minority state to majority
you need 2 unlikely
events in both cases





R Next recruitment will occur at time increment $\Delta t_R = -\log(\text{ran}_1)$ with uniform $\text{ran}_1 \in [0, 1]$.

D Next direct conversion event will occur at time increment $\Delta t_D = -\frac{\log(\text{ran}_2)}{\beta}$ with uniform $\text{ran}_1 \in [0, 1]$.



3 If $\Delta t_R < \Delta t_D$ one randomly select two different agents. If these two agents are both in R-state, or both in L-state one select a third agent. This third agent is now set to the same state as the two agents.

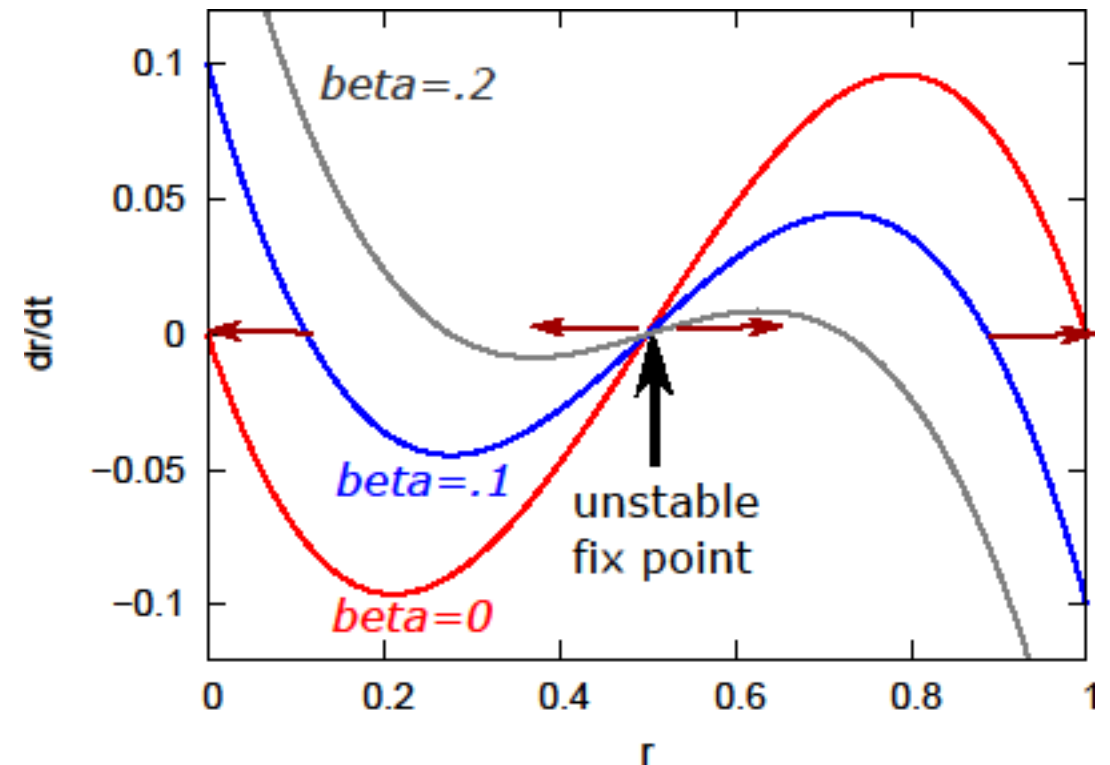
4 If $\Delta t_D < \Delta t_R$ one select one random agent and change the state of this agent to the opposing state.

5 Update time according to the selected event.

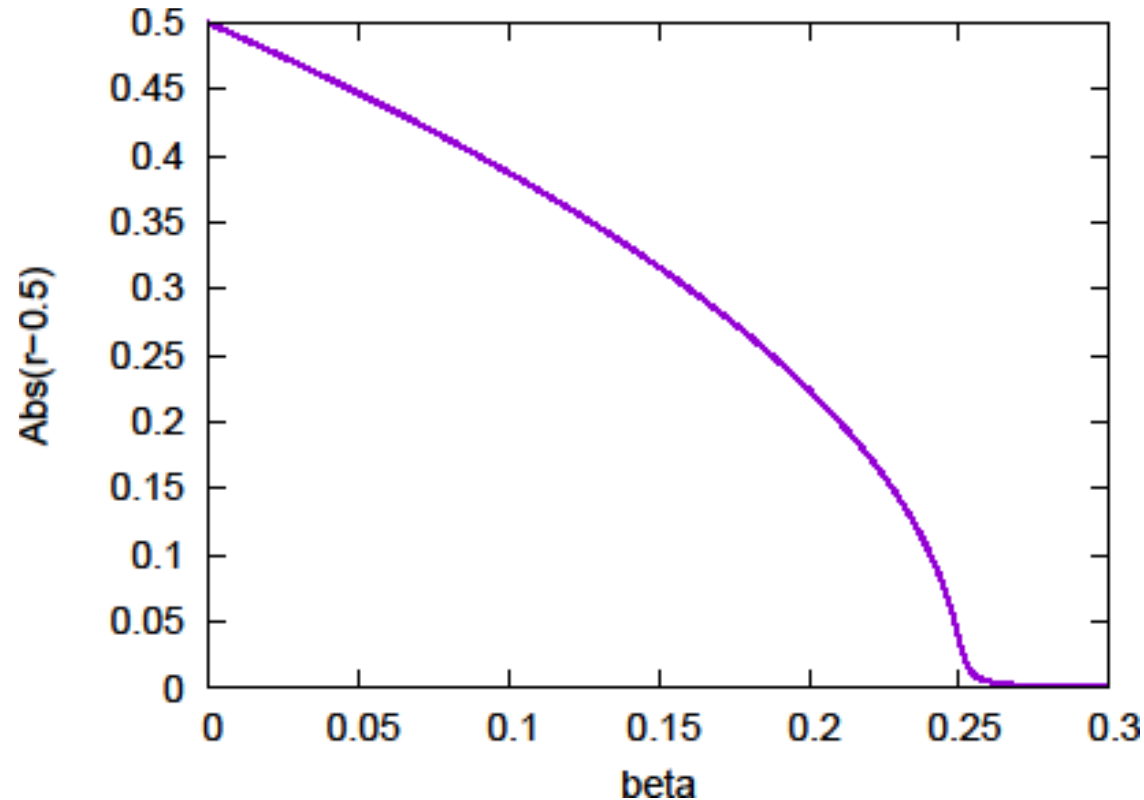
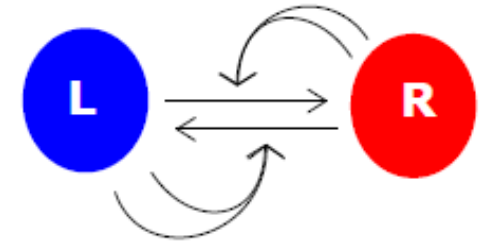
→ Social model where majority is more convincing:

$$\frac{dr}{dt} = (r^2 \cdot (1 - r) - r \cdot (1 - r)^2) - \beta \cdot r + \beta \cdot (1 - r) + \xi(t).$$

$$\begin{aligned}\frac{dr}{dt} &= r(1 - r)(2r - 1) + \beta \cdot (1 - 2r) + \xi \\ &= (r(1 - r) - \beta)(2r - 1) + \xi\end{aligned}$$



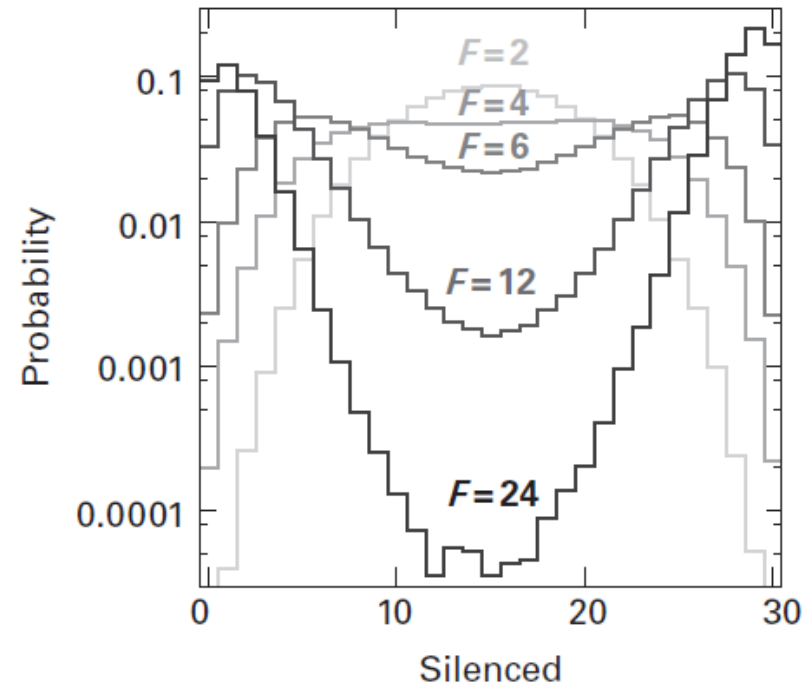
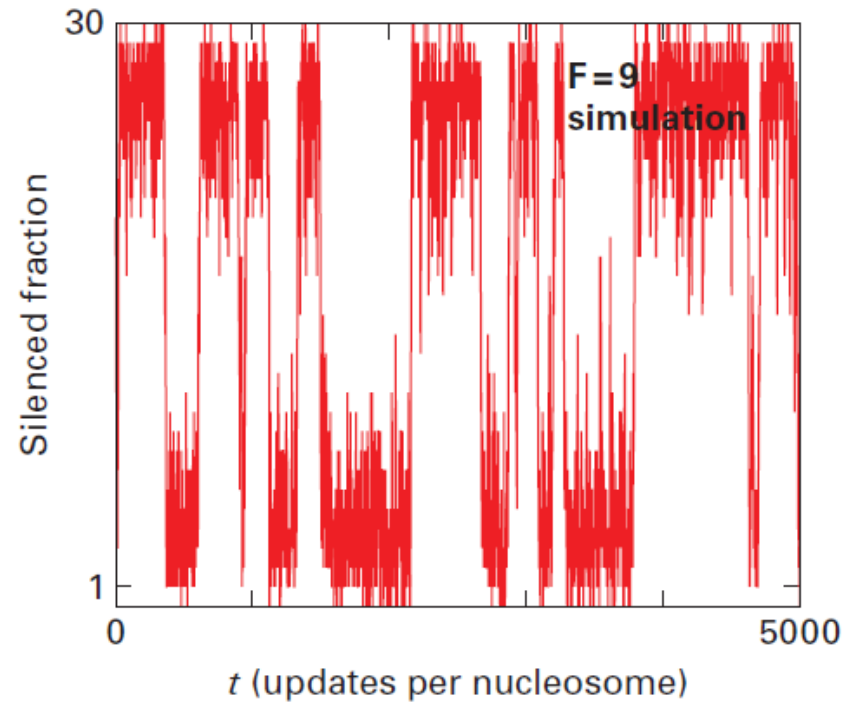
“Magnetization” in random neighbor version of



$$r = 1/2 \pm \sqrt{1/4 - \beta}.$$

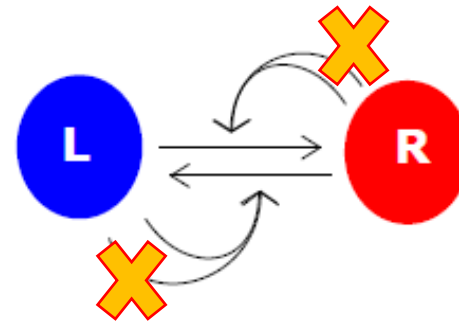
Noise value $\beta = 1/4$ is critical.
Exponent for magnetization
Below criticality,
(exponent unfortunately
also called β) is $1/2$
Just as in the Ising model.

2-state cooperative model for $N=30 \rightarrow$ Bi-stability:



$$F=1/\beta$$

Square term important:



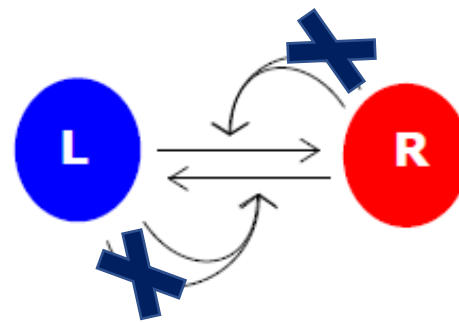
Without a square term, we just have a random neighbor voter model + some noise:

$$\begin{aligned}\frac{dr}{dt} &= (r(1-r) - r(1-r)) + \beta(1-2r) + \xi \\ &= \beta \cdot (1-2r) + \xi\end{aligned}$$

→ Always stable fixed point at $r=1/2$.

Mini tutorial: Give reasons for square versus linear term in opinion dynamic.
Give examples of all or none dynamics in human society?

Gillespie Simulation of:



No bi-stability

R Next recruitment will occur at time increment $\Delta t_R = -\log(\text{ran}_1)$ with uniform $\text{ran}_1 \in [0, 1]$.

D Next direct conversion event will occur at time increment $\Delta t_D = -\frac{\log(\text{ran}_2)}{\beta}$ with uniform $\text{ran}_1 \in [0, 1]$.

3 If $\Delta t_R < \Delta t_D$ one randomly select ~~two~~ **one agent** events. If the ~~selected~~ 

Select another agent and set it to the same state as the **first agent**.

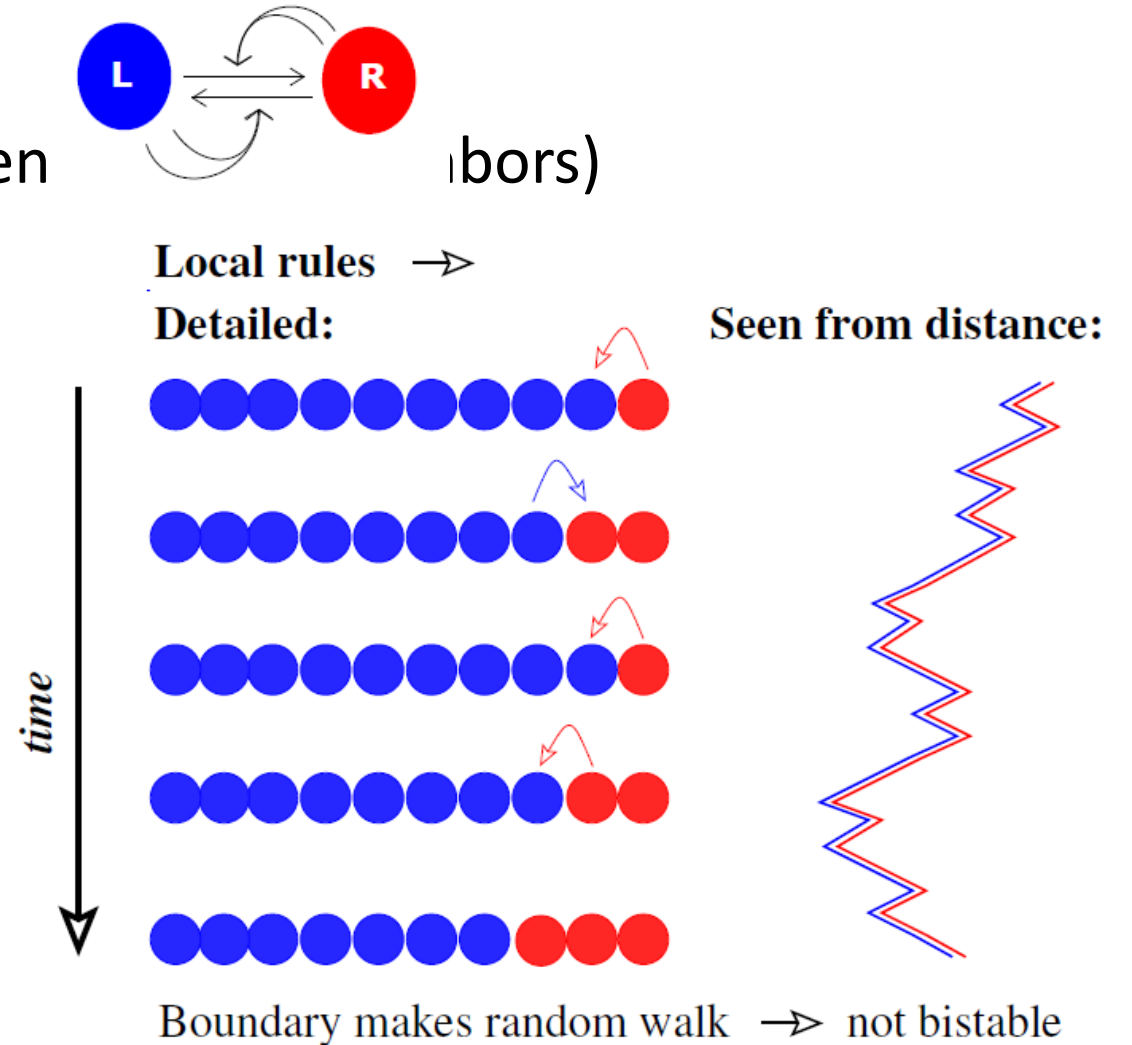
4 If $\Delta t_D < \Delta t_R$ one select one random agent and change the state of this agent to the opposing state.

5 Update time according to the selected event.

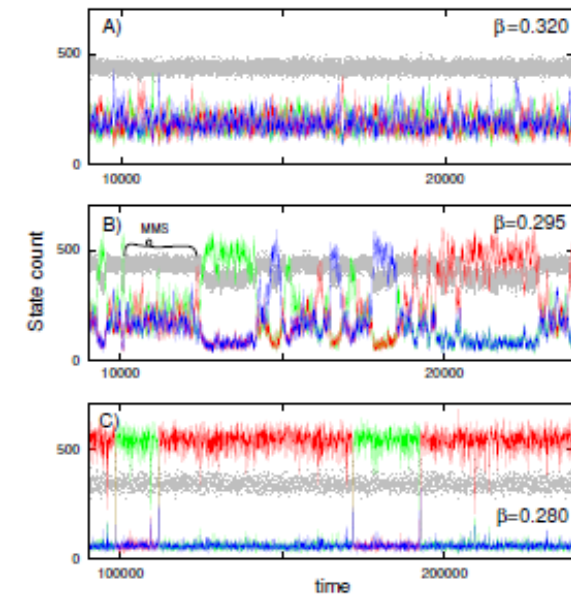
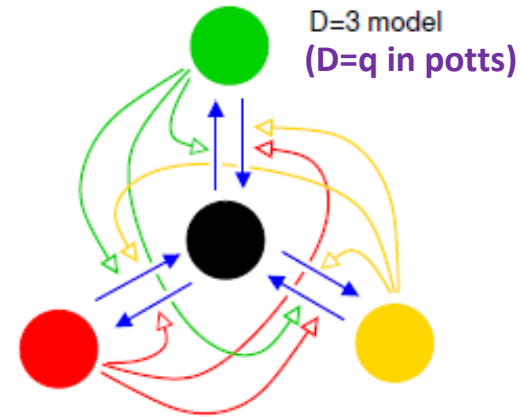
Also: No bi-stability in 1-dimension

(similar to No phase transition in 1 dimensional equilibrium models)

- Purely local rules,
- $\rightarrow$ have to have 2 neighbors (that then
- **Random walk** of boundary
- $\rightarrow$ no phase transition, i.e.
stability of any state is marginal:
Even complete **blue state** have chance **$\beta \cdot \beta$**
to do two neighbor conversions
To red, which in turn have chance
 $1/N$ to switch to pure **red** state.

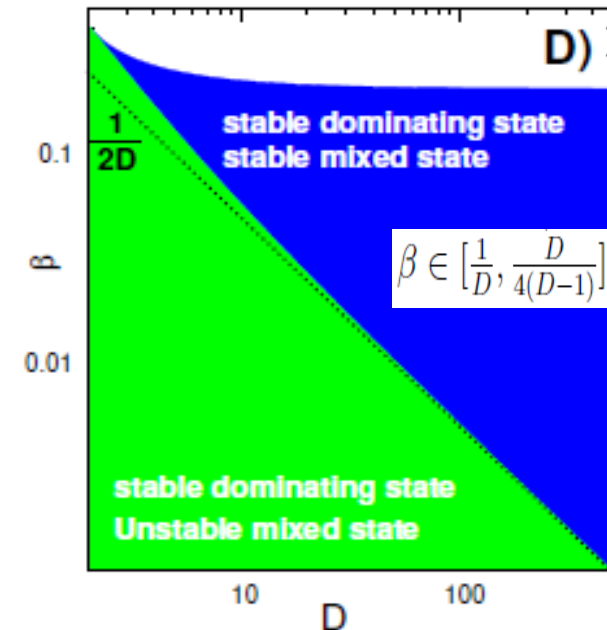
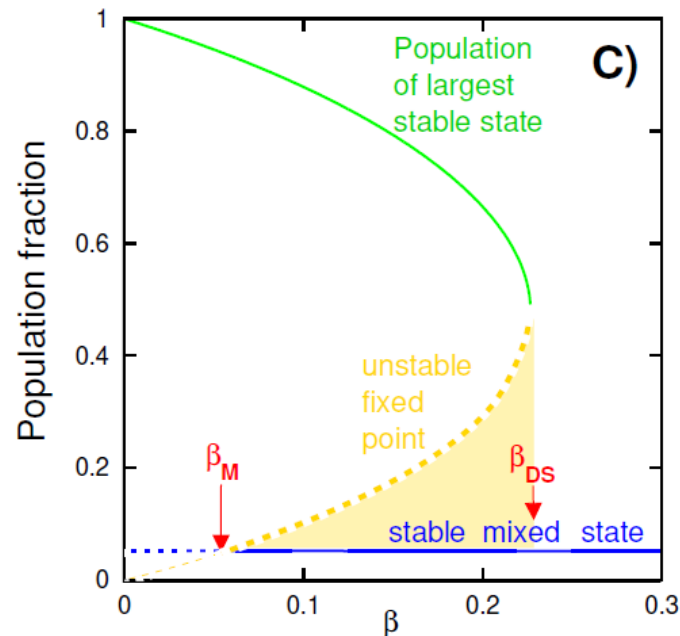
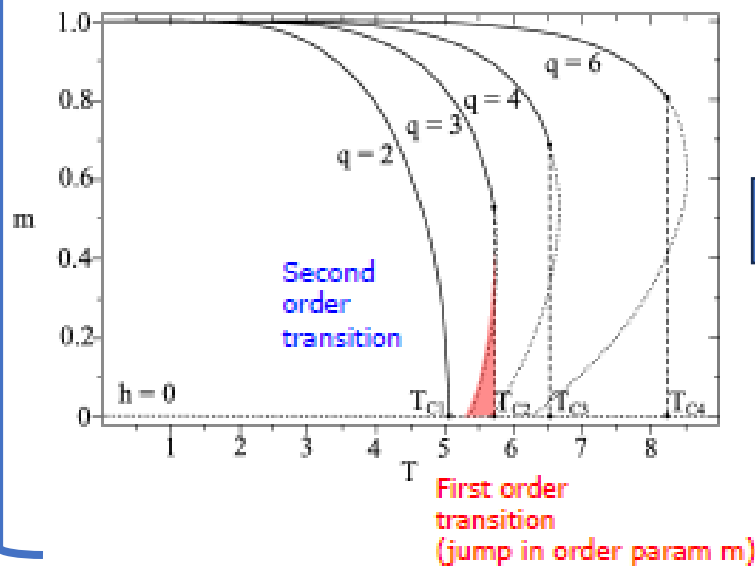


...and to a Potts model analog (see chapter 9):

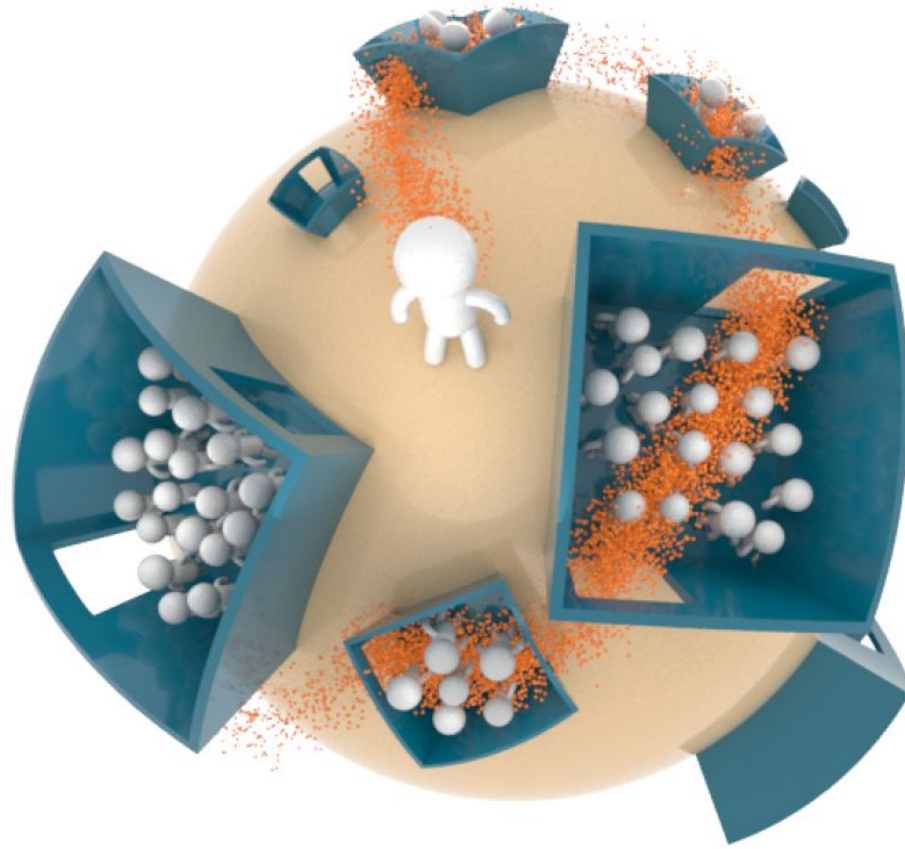


Here the shared state is supported by a barrier between $1/D$ to $1/2$;

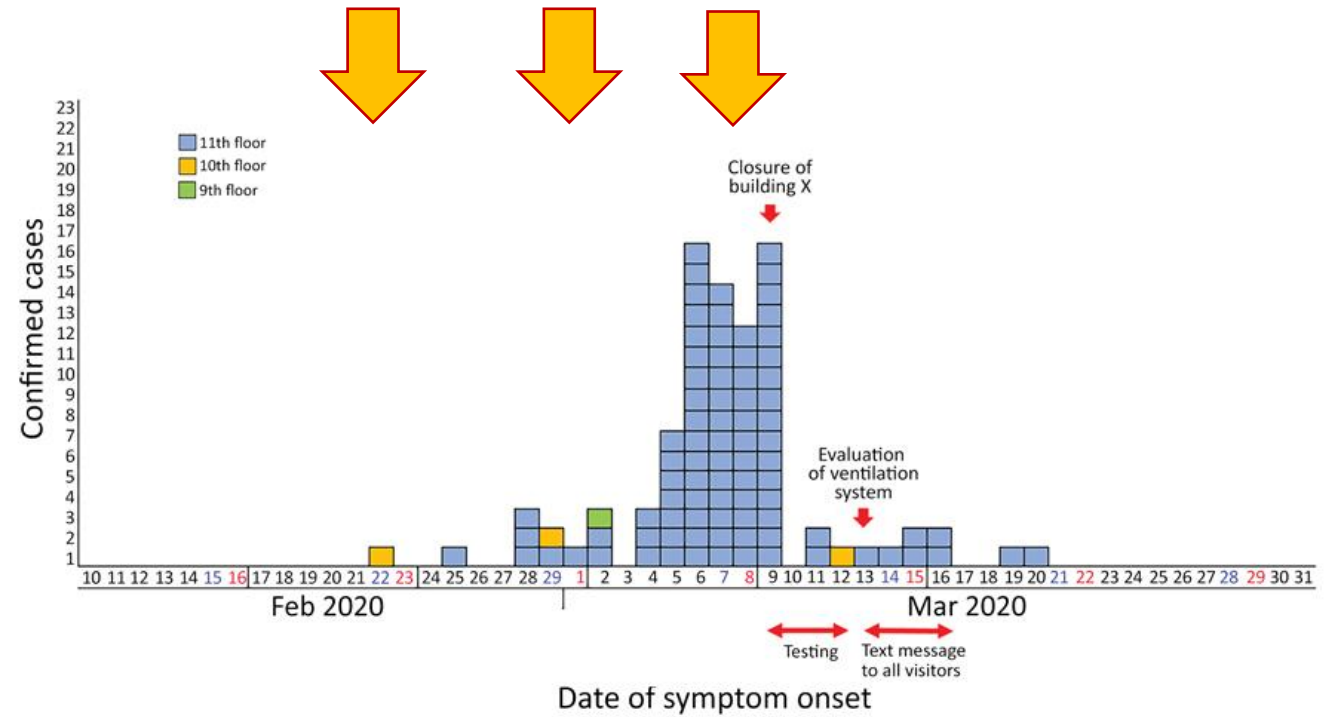
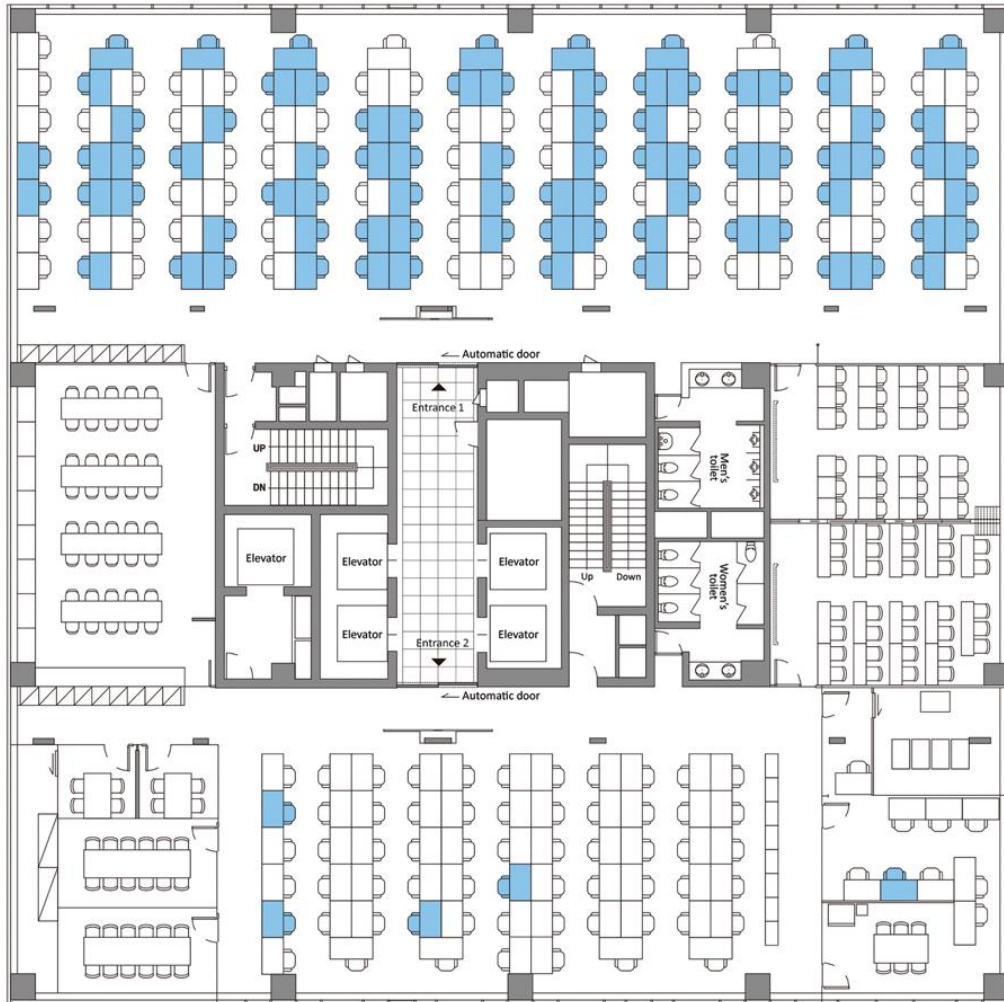
Random neighbor Potts model, different q :



Epidemics:



Korea Call center:... 1(?) $\rightarrow$ 7 $\rightarrow$ 68...(Lock-Down)



R0 of about 8, and timescale is 5 days

Super critical branching with $R_0=3$:

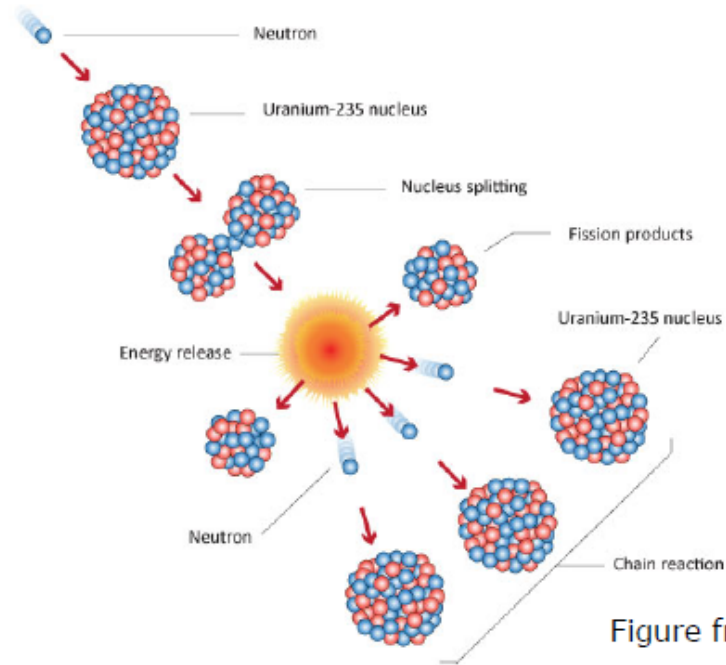


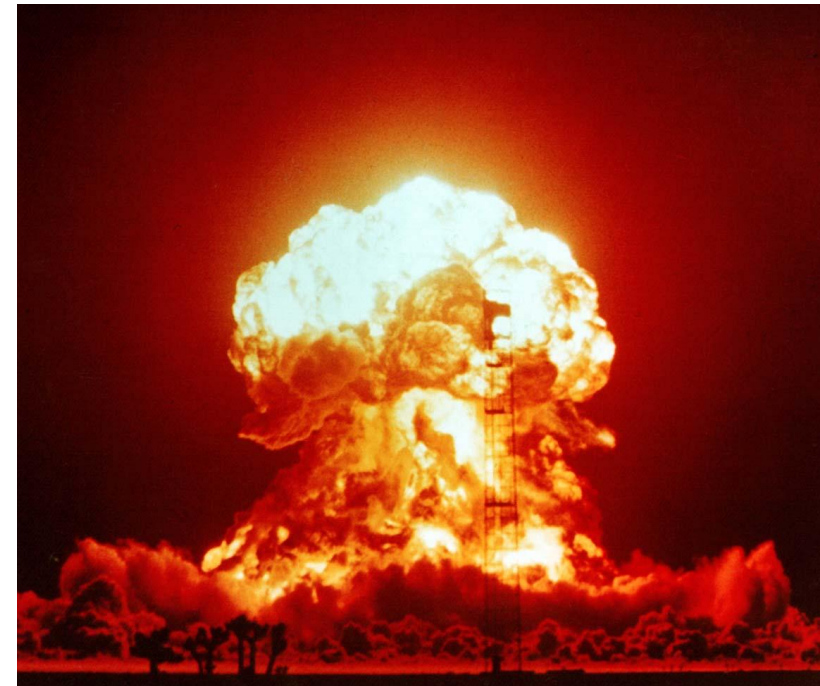
Figure from passmyexams.co.uk/GCSE/physics

Super-Critical chain reaction: each reaction give rise to > 1 new reaction

Critical chain reaction: each reaction give rise to one new reaction

Sub-Critical chain reaction: each reaction give rise to < 1 new reaction

....Another process with $R_0 = 3$
....somewhat faster timescale



In reality $S \rightarrow E \rightarrow I \rightarrow R$ with fairly well defined length of states:

- Residence in e.g. is not purely exponential, as it takes time for virus to replicate before infection possible:

$$P_1(t) = \frac{1}{\tau} \exp(-t/\tau)$$

$$P_2(t) = \int_0^t P_1(t') \cdot P_1(t - t') dt' = \frac{t}{\tau^2} \exp(-t/\tau)$$

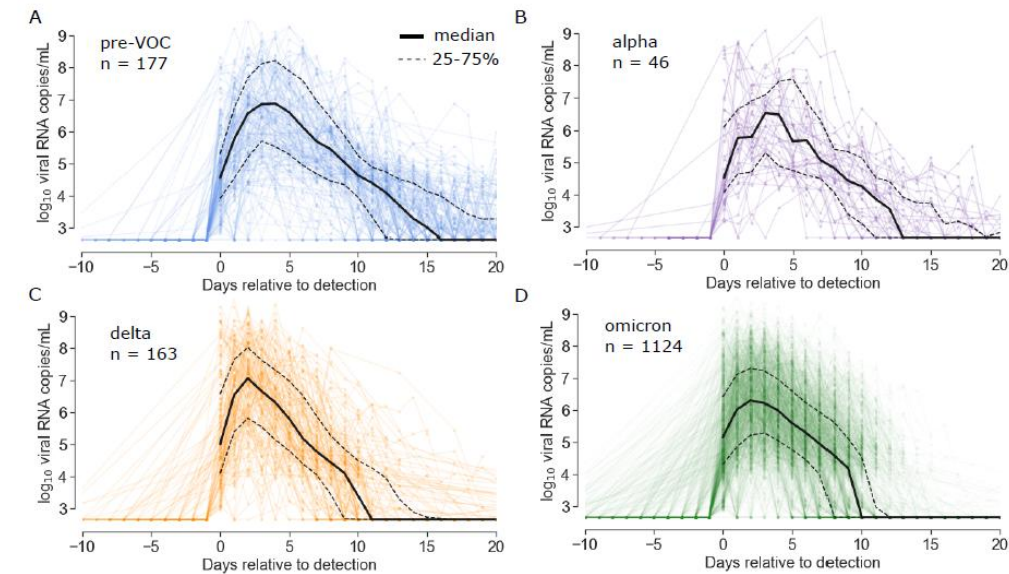
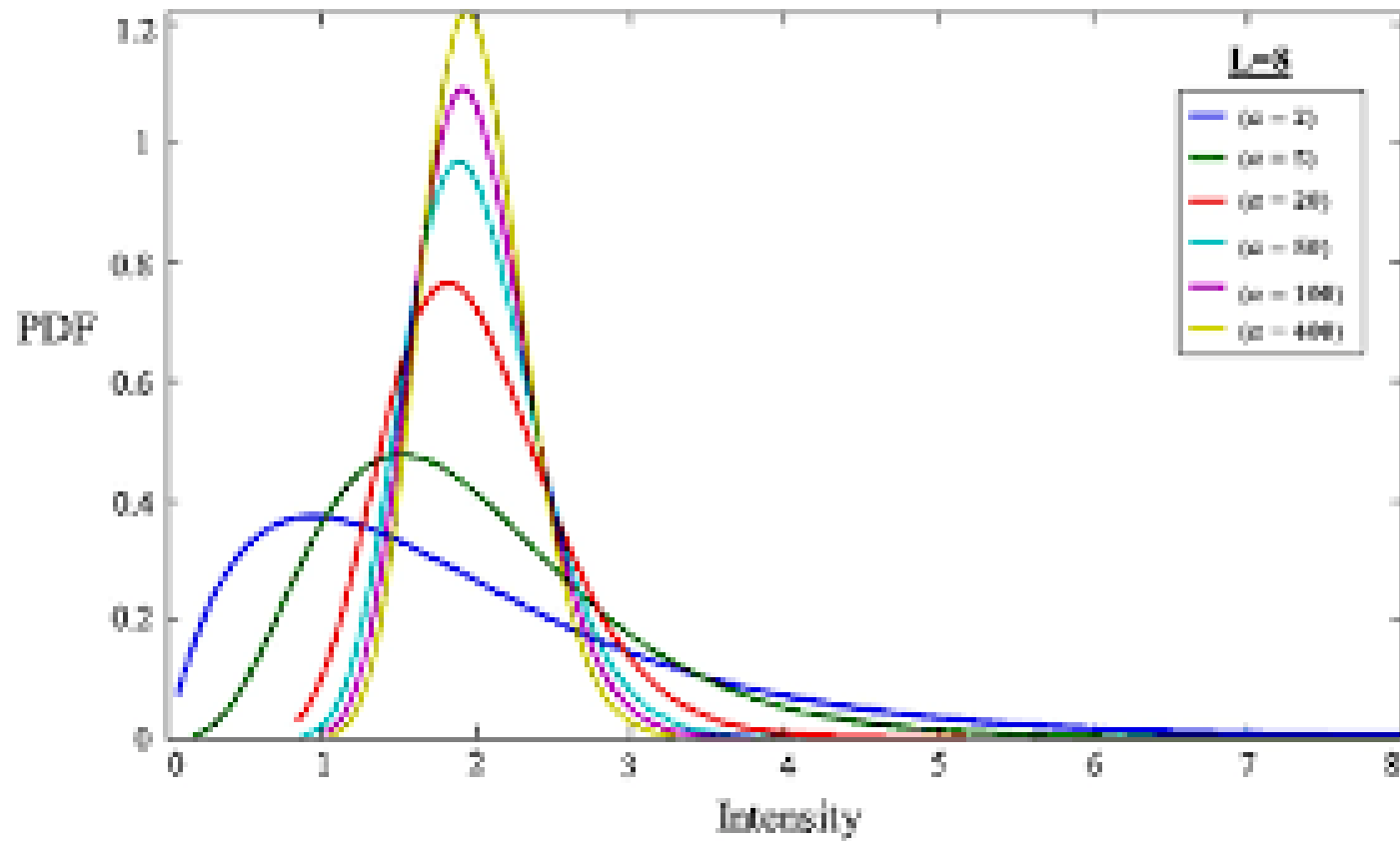
$$P_3(t) = \int_0^t P_2(t') \cdot P_1(t - t') dt' \propto \frac{t^2}{\tau^3} \exp(-t/\tau)$$

From exponential
Towards a more peaked
Distribution with increasing
number of steps.

$$\begin{aligned} \dots \quad \frac{dP_1}{dt} &= \text{input}(t=0) - I_1/\tau \\ \frac{dP_2}{dt} &= I_1/\tau - I_2/\tau \\ \frac{dP_3}{dt} &= I_2/\tau - I_3/\tau \end{aligned}$$

Can be represented by Gamma functions

Gamma distributions for $k=2$ to $k=400$



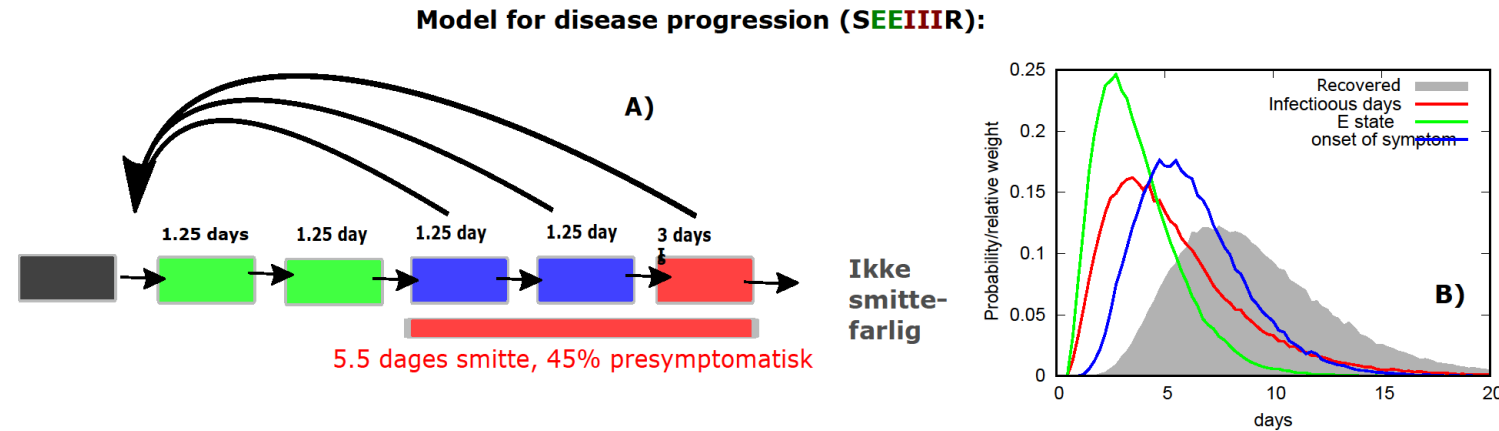
Example: Model for covid disease progression → timing of tests

Follow an individual
after a potential infection

sample individuals and count
when they infect. Then attempt to
test at specific days and assume
person quarantined after first
positive test.

find optimal days to test:
→ test on day
4 And on day 6.

Logic: We follow a sick from infection time and simulate how many days he infect
This number of days should be minimized by inserting tests and removal of positives..
Persons are infectious before test, but quarantined afetr first positive test.



1) One test should be on day 5 which will reduce infection
from 5days to **2.7 days**

2) Two tests: Test on day 4 and day 7
reduce infection from 5 to **1.7 infection days**

20% for false negative: test on day 5 → **3.2 days**.

20% for for false negative; Test on day 4,6 → **2.2 smittedag**.

$$\begin{aligned}\frac{dP_1}{dt} &= \text{input}(t=0) - I_1/\tau \\ \frac{dP_2}{dt} &= I_1/\tau - I_2/\tau \\ \frac{dP_3}{dt} &= I_2/\tau - I_3/\tau\end{aligned}$$

Epidemics, SEIR $\equiv$ SIR models:

- Random encounters:

Frequency for an encounter between A and B $\propto \rho_A \times \rho_B$

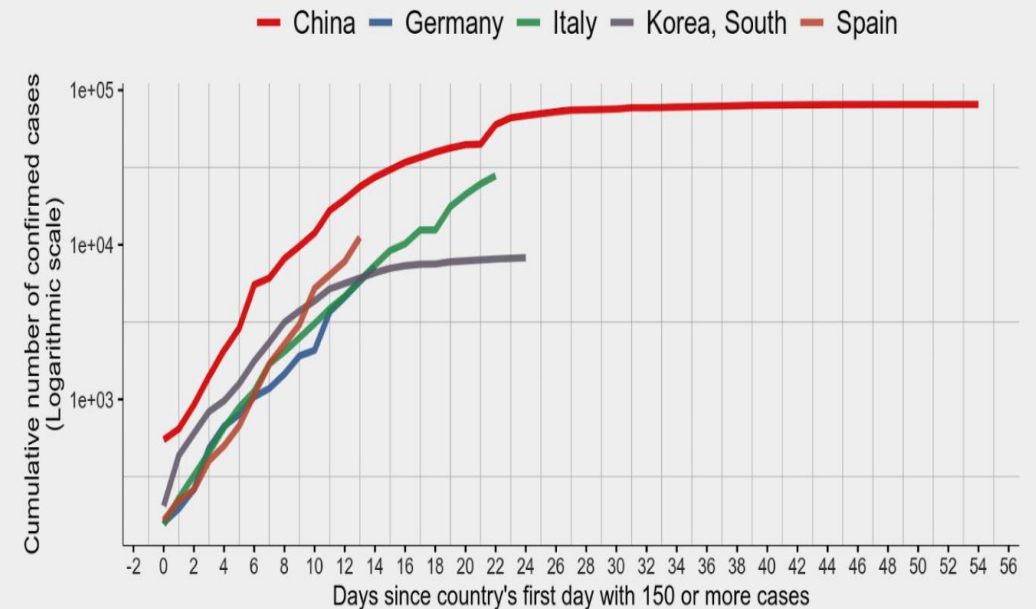
$$\begin{aligned}\frac{dS}{dt} &= -\lambda \cdot S \cdot I \\ \frac{dI}{dt} &= \lambda \cdot S \cdot I - \gamma I \\ \frac{dR}{dt} &= \gamma \cdot I\end{aligned}$$

→ exponential growth



COVID-19 cases since country's
first day with 150 or more cumulative cases

Data as of 2020-03-16



SIR models:

$$\begin{aligned}\frac{dS}{dt} &= -\lambda \cdot S \cdot I \\ \frac{dI}{dt} &= \lambda \cdot S \cdot I - \gamma I \\ \frac{dR}{dt} &= \gamma \cdot I\end{aligned}$$



$$R_0 = \frac{\lambda}{\gamma},$$

And infecting R_0 people takes $1/\gamma$ time

Growth rate=

$$dI/dt = (\lambda - \gamma) I \dots \text{ (for } S \sim 1)$$

Herd immunity: This is level of immunization to stop I from growing.

Herd immunity ($dl/dt=0$), q fraction of people which are immune ($S=1-q$):

$$R_0 \cdot (1 - q) < 1 \rightarrow q > 1 - \frac{1}{R_0}$$

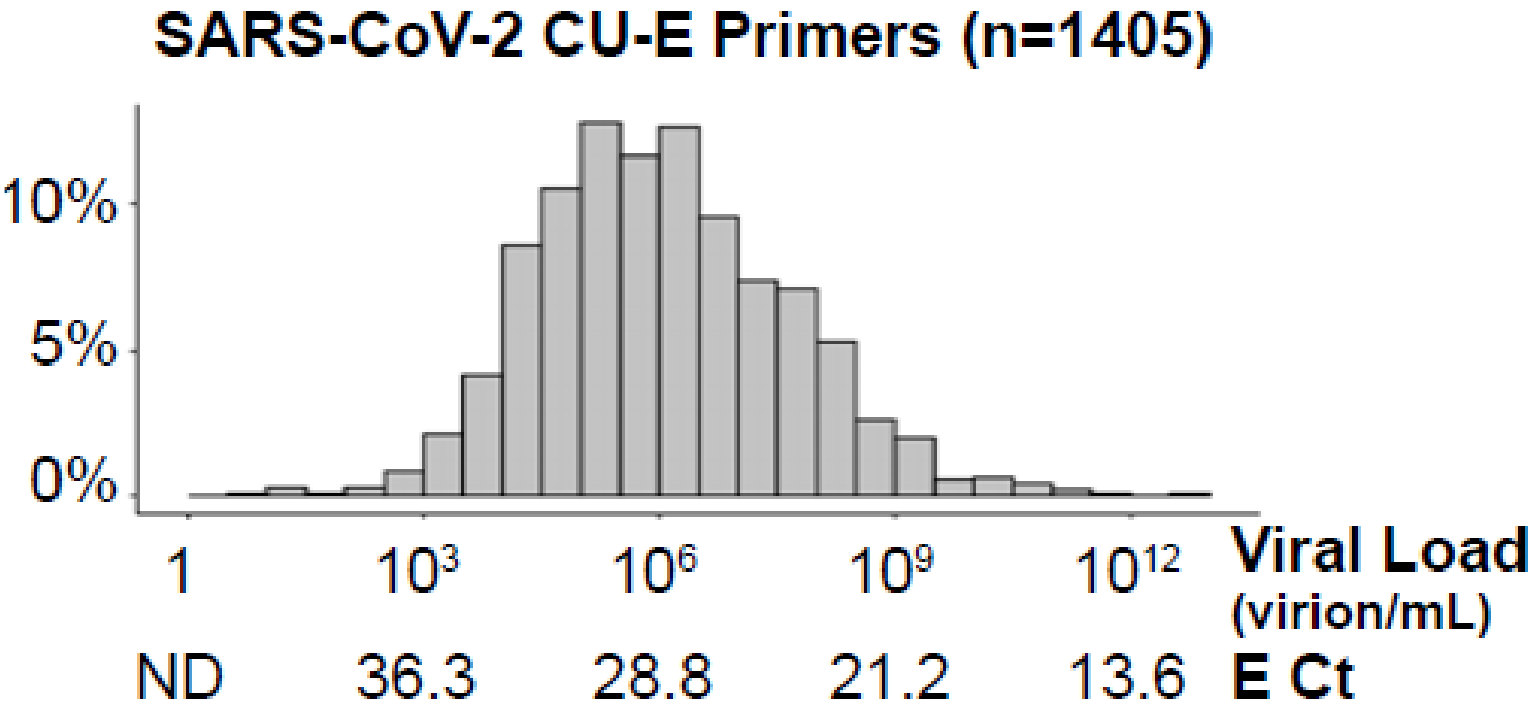
Total number of infected people if we let epidemic grow unmitigated:

$$\frac{d \ln(S)}{dR} = -\lambda/\gamma = -R_0 \Rightarrow S(\infty) = e^{-R_0(1-S(\infty))}$$

For R_0 of order 3 or above then $S=\exp(-3) \ll S=1/3$

... mitigation is a good idea, else everybody gets infected in the end

**Super shedders in the Covid-19 epidemic:
2% of people carry 90% of the virus:**



Tutorial in time use: 5 days infection period for Covid-19.

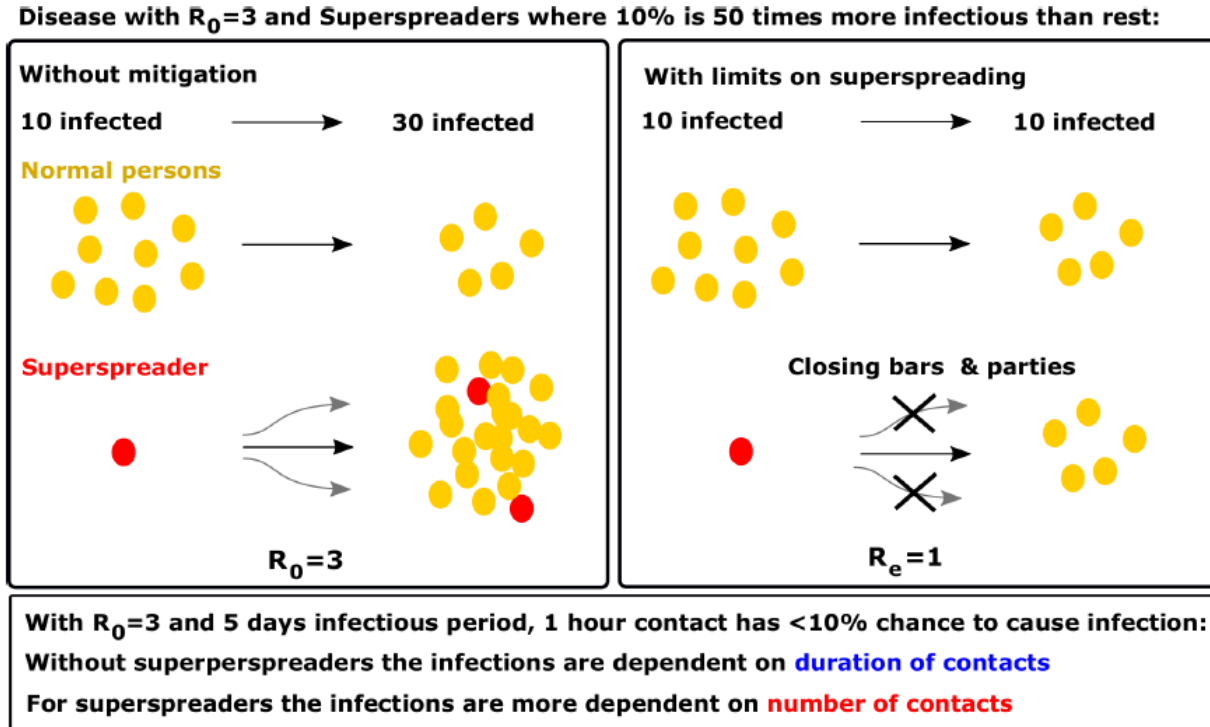
$R_0=3$ & people are social say 12 hours a day. This means 3 infections per 60 hours
Or 5%infection probability per hour when together.

E.g. it have been reported that sitting besides Covid-19 patient in train in China
gives risk of 1.2% per hour. And sitting on the seat after person have left is 0.1% risk.

→thus it doesn't matter much if you spent time with different persons each hour, because
risk of infection is additive.

Superspreaders in Covid-19

- An example where Agent based models becomes crucial.

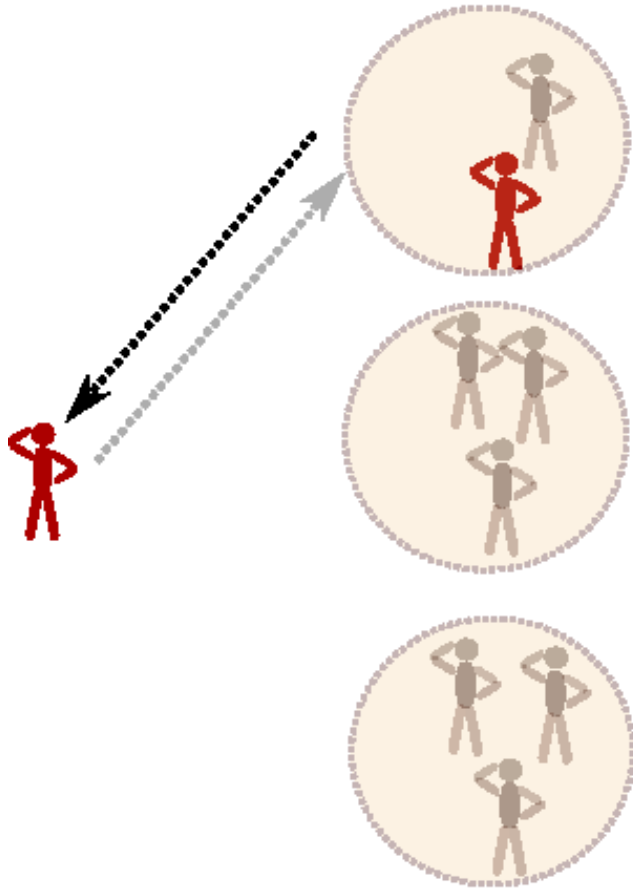


Have to deal with confines of social Networks, and Finite time

Disease spreading needs Infectability and Opportunity

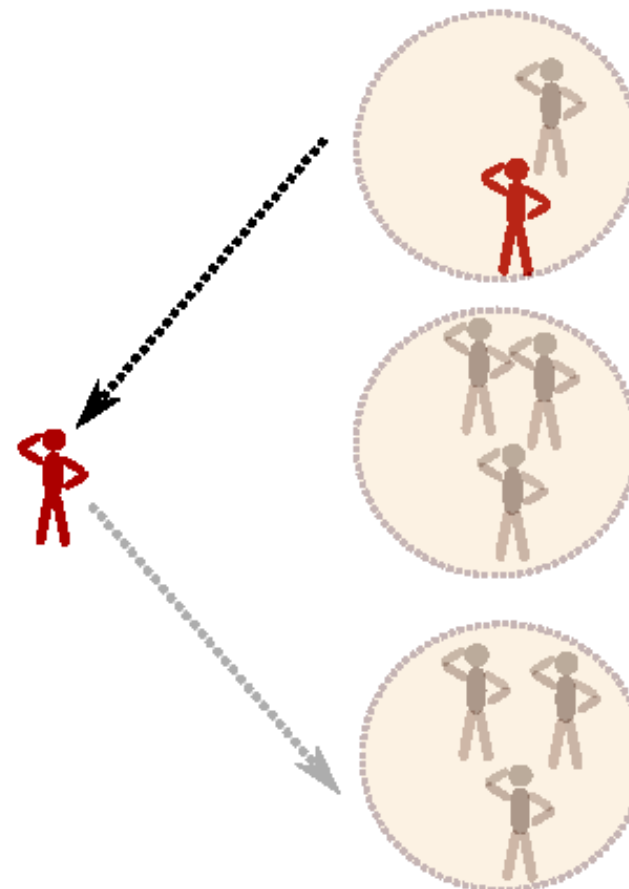
The latter nearly unlimited in SIR / "mean field" models

Reality: Each man return to same home every day



Here can only infect up to 2 new susceptibles

SIR simplification: Each man return to a new home every day

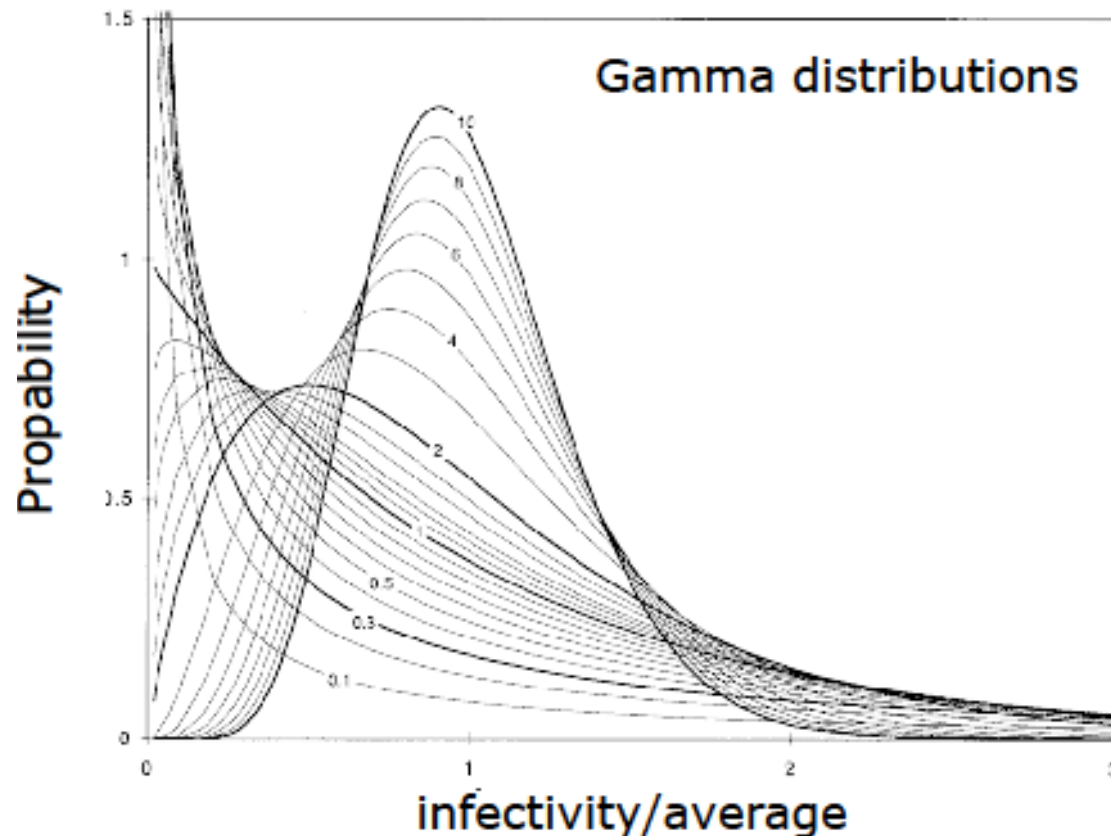


Here he has unlimited infection potential (one for each timestep)

Gamma distributed infectivity:

$$P_I(r) \propto r^{k-1} \cdot \exp(-kr/\mu)$$

μ represent the average.



Advantage of Gamma over power law:

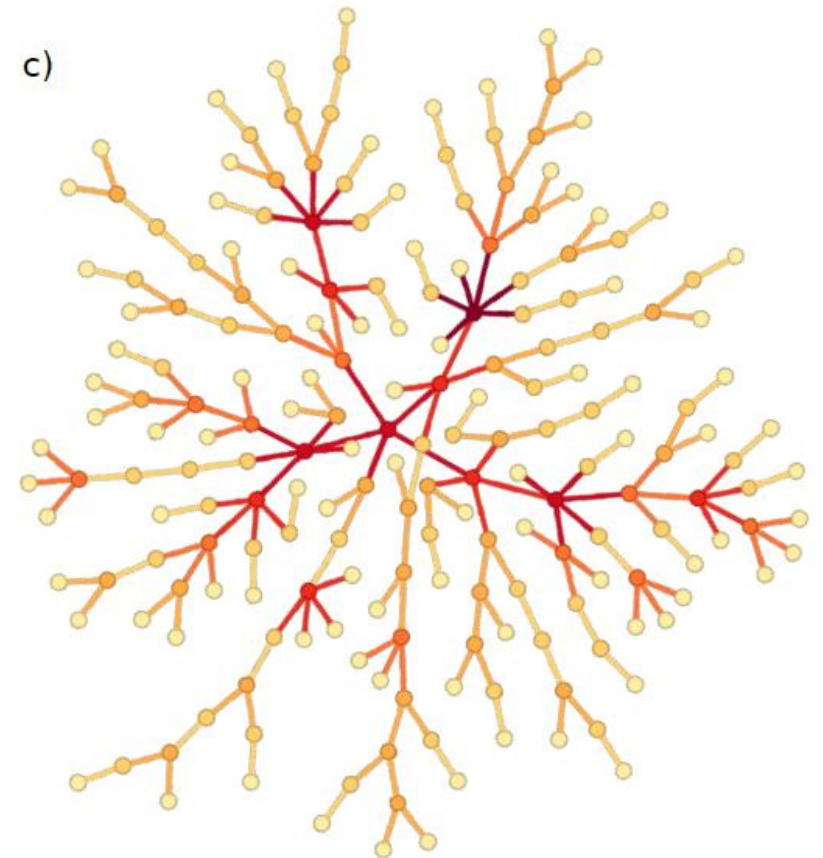
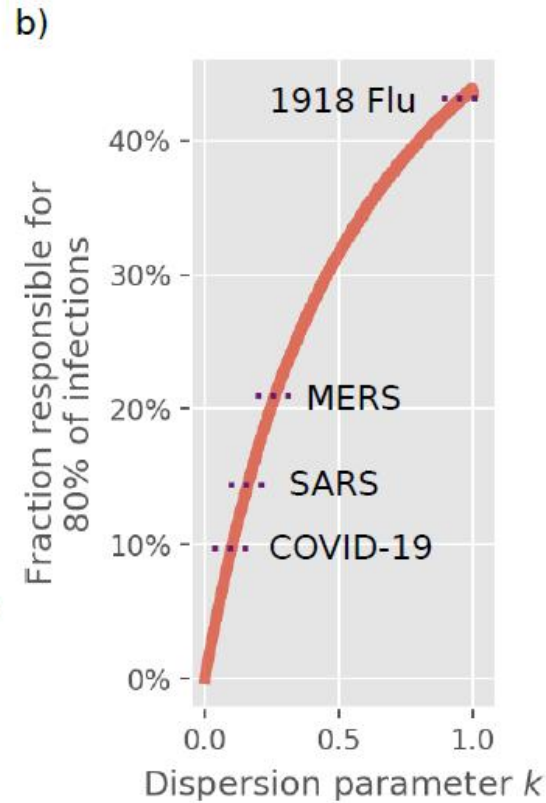
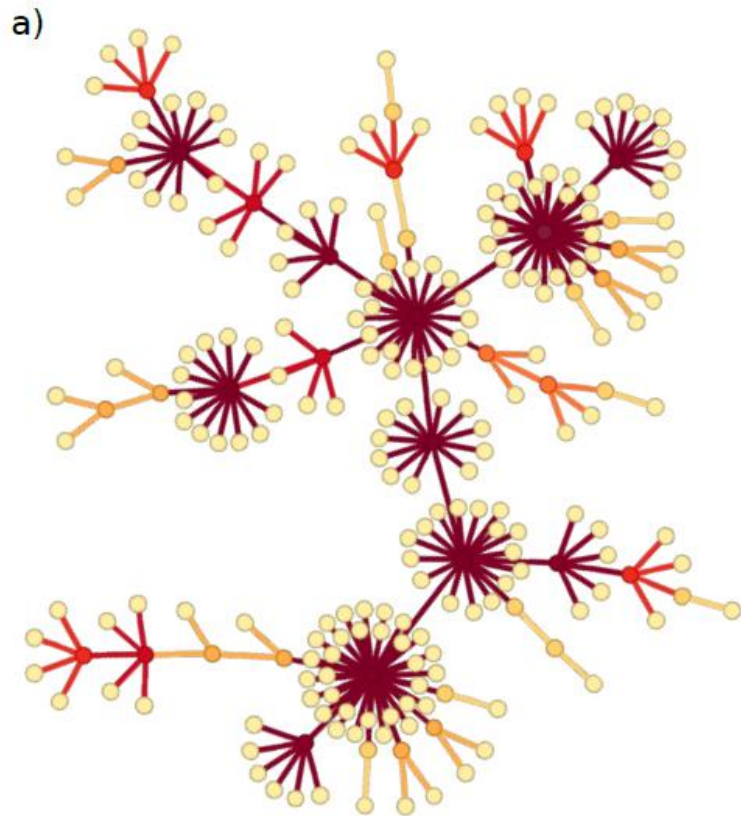
Only one parameter, and this allow
Variance to grow indefinitely ($=1/k$)

In power law we would
need power between 2 and 3
To obtain this (for flatter power laws
we need upper cut off & exponent).

Anyway a Gamma distribution is divergent
As a power law as we approached 0.

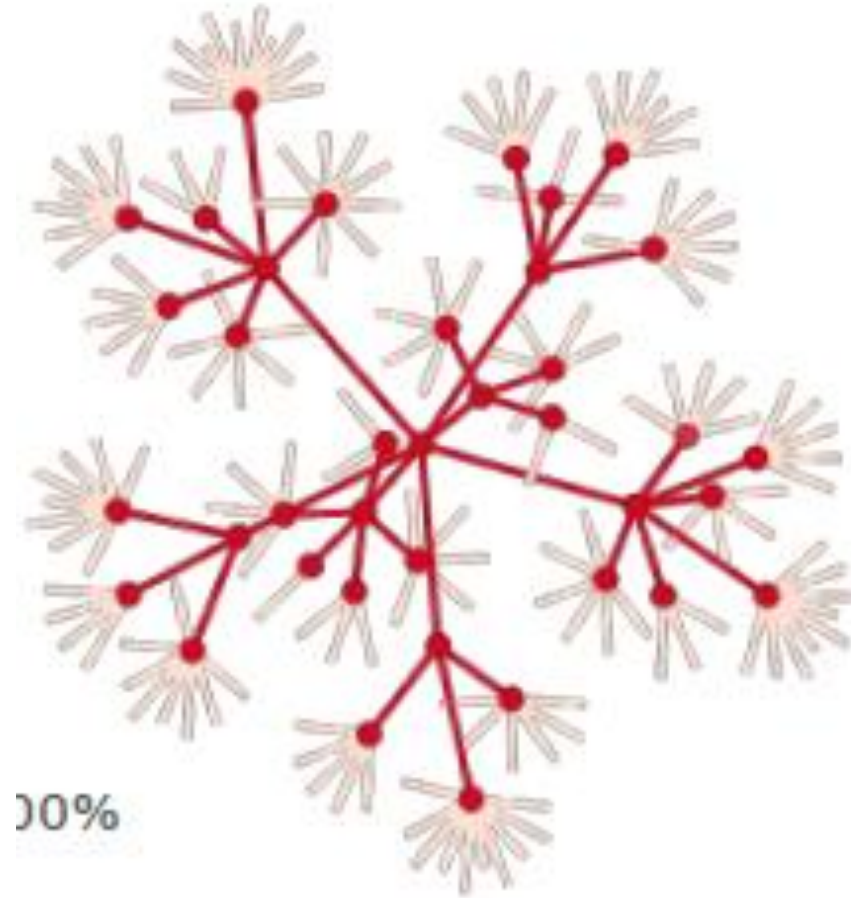
Gamma distribution infectivity was
introduced by Lloyd in 2005.

Infection pattern with Gamma distributed infectivity

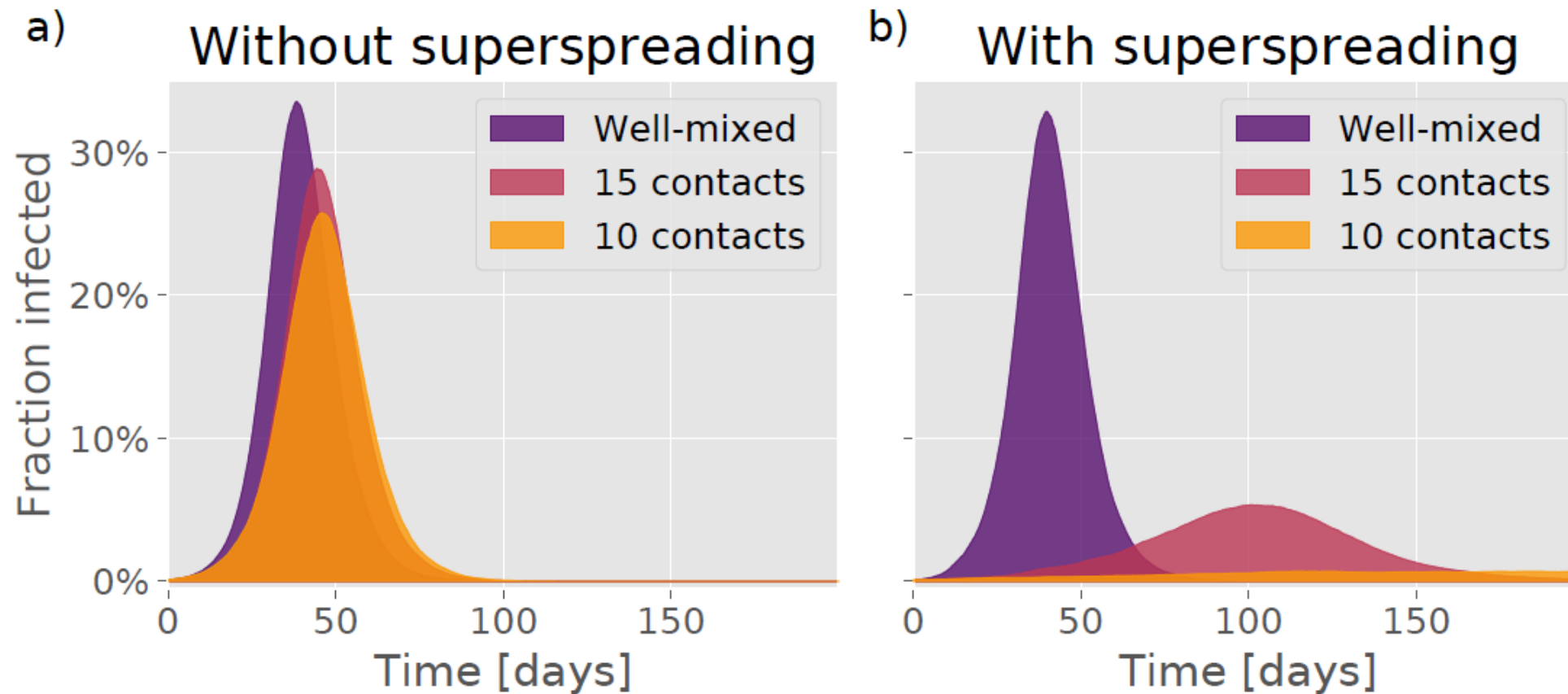


Agent-based model

(Quenched contacts for individuals, ER network)

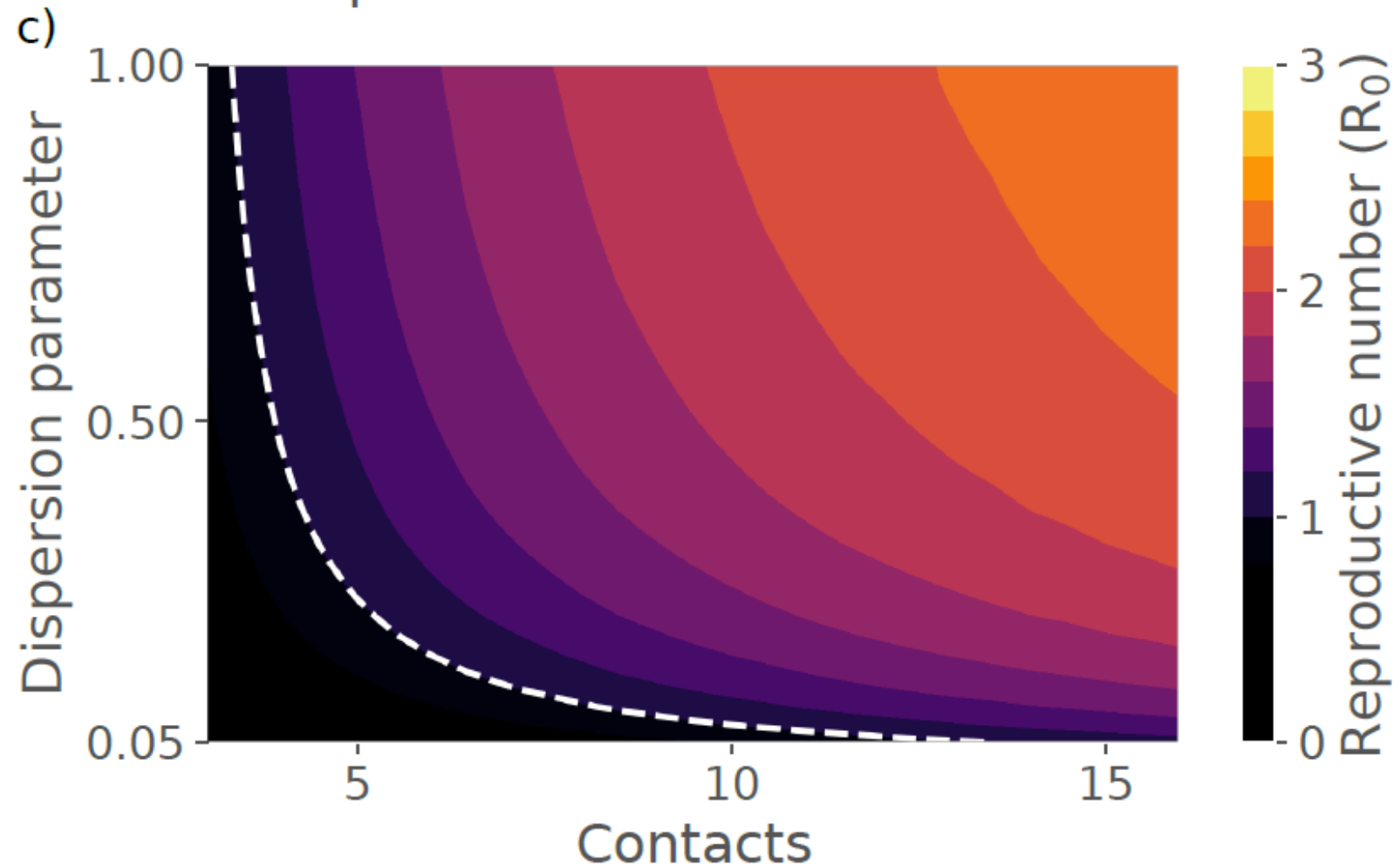


Running an epidemic on a size 10000000 ER network
at different average connectivity
(basic free R_0 is 3, like in classic Covid-19):

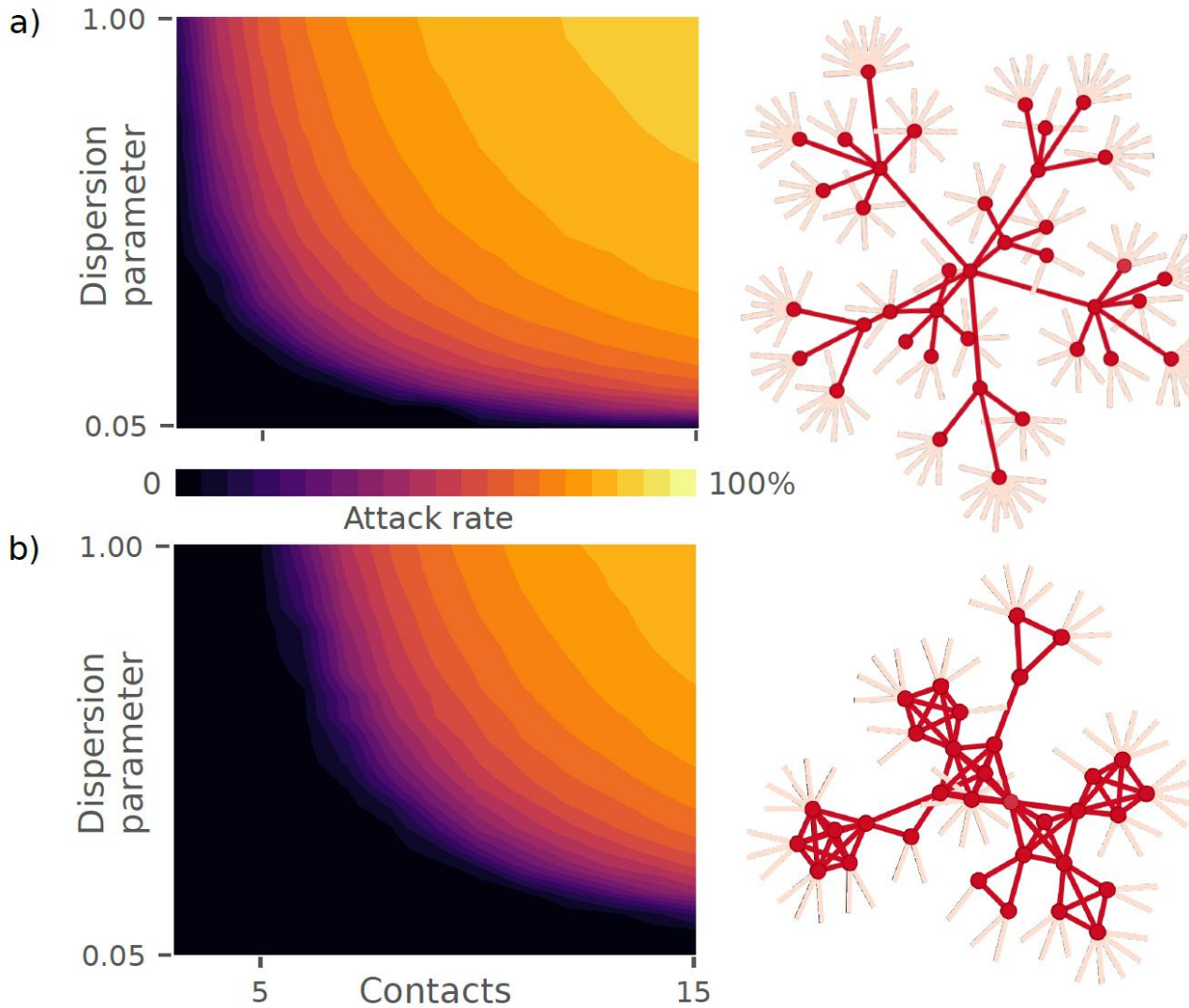


Epidemic stop when c below 14!

Average over all individuals:
Epidemic stop in black area



Mitigation depend on network clusterization:



The attack rate of the disease is clearly lowered by clustering in general (Fig. 4.28), but the effect is especially significant when heterogeneity is high. The mechanism behind this is that of *local saturation*. If a superspreader infects a significant portion of his network, there is a risk that one of these individuals will turn out to be another superspreader. However, *if* there is clustering, a large part of this *second* superspreader's network will already have been exposed, and the second superspreader does comparatively little harm.

Heterogeneous $\leftrightarrow$ Mean Field

